

Portfolio Theory Exam

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Introduction

In this work, we address a portfolio asset allocation problem based on two samples of US-listed equities. The analysis is organised into two main parts. The first establishes the statistical foundations and develops the core mean–variance optimisation for two samples of US-listed equities: 79 securities from the Nasdaq 100 and 129 securities from the New York Stock Exchange. It begins by characterising their distributional properties at daily and monthly frequency, constructs variance–covariance and correlation matrices, and selects twelve representative securities from each exchange. These securities are then used to build unconstrained and long-only tangency portfolios, benchmarked against the Nasdaq Composite and S&P 500 indices.

The Nasdaq sample is mainly composed of high-growth technology, semiconductor and software firms, whereas the NYSE sample is more diversified, including industrials, energy, financials, health-care and consumer staples. Although Nasdaq portfolios deliver higher absolute returns, NYSE portfolios generally achieve superior risk-adjusted performance across model specifications and frequencies.

The second part extends the analysis in two directions. First, the estimated portfolios are linked to the Capital Asset Pricing Model through beta estimation and Security Market Line verification. The results show that securities and portfolios in both markets tend to lie above the Security Market Line at both frequencies, producing positive alphas across different estimation windows. This suggests that the single-factor CAPM is not sufficient to explain realised returns over the sample period.

Second, the report considers alternative portfolio construction methods designed to address the limits of standard mean–variance optimisation. These include the Black–Litterman model, which uses a CAPM equilibrium prior, the Global Minimum Variance Portfolio, which relies only on the covariance matrix, and pure Bayesian approaches. The final stage combines mean–variance, Black–Litterman, GMVP and Bayesian portfolios through linear weighting schemes ranging from risk minimisation to return maximisation.

A consistent empirical pattern emerges: monthly returns generally provide better risk-adjusted performance than daily returns, while long-only portfolios often match or outperform unconstrained ones in Sharpe ratio, since short selling may reduce expected return more than variance. Among the combined portfolios, equal weighting across models provides a robust benchmark.

All prices are adjusted for total returns, including dividend reinvestment. Returns are computed at daily frequency, using 1305 observations from early 2021 to early 2026, and monthly frequency, using 60 observations over the same period. The two frequencies serve as a robustness check for portfolio weights, Sharpe ratios, beta estimates and method rankings. The risk-free rate is set at $r_f = 2\%$ per annum and converted to the corresponding daily or monthly rate when required.

1-2. Summary Statistics

Nasdaq

At daily frequency, mean returns across the 79 Nasdaq stocks range from -0.000115 for Astrotech to 0.003844 for Quantum Computing, with a cross-sectional average of 0.000583 per day and a median of 0.000540 . Daily standard deviations span from 1.13% for Roper Technologies to 8.33% for Quantum Computing, with an average of 2.48% , confirming the wide dispersion in risk levels even within a single exchange. At monthly frequency, mean returns range from -3.37% for Astrotech to 8.83% for Quantum Computing, with an average of 1.25% and a median of 1.16% . Monthly standard deviations vary from 5.18% to 53.02% , with a cross-sectional average of 11.68% .

Skewness at daily frequency has a mean of 0.347 , with 48 positively skewed and 31 negatively skewed stocks. At monthly frequency, mean skewness rises to 0.502 and positively skewed stocks increase to 57 out of 79. Excess kurtosis at daily frequency averages 14.81 , far above the Gaussian benchmark of zero. Extreme cases include Astrotech with 182.2 , Biogen with 112.5 , Abvc Biopharma with 66.6 and Vimeo with 54.7 . These values indicate that extreme observations occur far more frequently than predicted by a normal distribution, implying genuinely heavy tails rather than simple volatility. At monthly frequency, average excess kurtosis falls sharply to 1.61 , confirming that temporal aggregation reduces the impact of extreme observations and makes the Gaussian approximation more defensible for portfolio construction purposes.

The average annualised Sharpe ratio for the Nasdaq universe is 0.301 at daily frequency and 0.297 at monthly frequency.

NYSE

For the NYSE sample, mean daily returns across the 129 stocks range from -0.001109 for Boston Beer to 0.002154 for Carpenter Technology, with an average of 0.000336 and a median of 0.000323 . Both figures are materially lower than the corresponding Nasdaq values, reflecting the less growth-oriented nature of the NYSE universe. Daily standard deviations range from 0.99% for Consolidated Edison to 6.12% for Alcoa, with an average of 2.05% , again below the Nasdaq average.

At monthly frequency, mean returns range from -2.42% for Boston Beer to 4.70% for Carpenter Technology, with an average of 0.71% and a median of 0.73% . Monthly standard deviations span from 4.68% to 31.49% , with an average of 9.53% , also lower than the Nasdaq monthly average of 11.68% .

Skewness at daily frequency has a mean of 0.152 , with 61 positively skewed and 68 negatively skewed stocks, showing a more symmetrical distribution than the Nasdaq sample. The most negatively skewed securities are Dxc Technology with -1.46 , Estee Lauder with -1.26 and Nike with -0.53 . At monthly frequency, skewness rises to 0.270 and positively skewed stocks increase to 93 out of 129.

Excess kurtosis at daily frequency averages 11.91 across the NYSE sample. There is one extreme outlier in Telephone & Data Systems with 208.8 , while other notable cases include Boston Beer with 21.6 , Dxc Technology with 15.1 and Nike with 14.6 . At monthly frequency, average excess kurtosis decreases substantially to 0.76 . The average annualised Sharpe ratio for the NYSE universe is 0.212 at daily frequency and 0.207 at monthly frequency, both substantially below the Nasdaq averages.

Comparison

The comparison between daily and monthly statistics highlights the importance of the time horizon in portfolio construction. Daily returns contain more short-term noise, temporary price reactions and isolated shocks, which explains the higher excess kurtosis and more pronounced tail risk. Monthly returns aggregate several daily observations into a longer interval, smoothing part of this noise and producing distributions that are generally closer to normality.

This distinction is particularly important for the mean–variance framework since it relies on the stability of expected returns and covariance estimates. At daily frequency, estimates are more sensitive to outliers and sudden market movements. Monthly estimates are generally less noisy, although the smaller number of observations reduces statistical precision. Using both frequencies therefore provides a useful robustness check: if the same securities or portfolios perform consistently under both daily and monthly data, the conclusions become more reliable.

Comment

Skewness and excess kurtosis are especially informative in this context. Positive skewness indicates the presence of large positive returns, which investors generally value because upside surprises improve portfolio performance. Negative skewness instead signals downside asymmetry and the possibility of losses larger than those suggested by volatility alone. Excess kurtosis measures the frequency of extreme observations relative to the normal distribution. High kurtosis therefore implies fat tails and a greater probability of extreme gains or losses, meaning that stocks with very high kurtosis should be interpreted carefully even when their average returns appear attractive.

Tables ?? and ?? report the individual statistics for a representative subset of six Nasdaq stocks, while Tables ?? and ?? present the same analysis for the NYSE sample. The tables report mean, variance, standard deviation, skewness, excess kurtosis and annualised Sharpe ratio at both frequencies.

Within the Nasdaq sample, Nvidia and Broadcom stand out because they combine high average returns with strong Sharpe ratios despite their elevated volatility. Pepsico and Comcast A instead display weaker risk-adjusted performance. In the NYSE sample, Carpenter Technology and Eli Lilly achieve the strongest Sharpe ratios, whereas Pfizer and 3M show weak or negative risk-adjusted performance. More generally, NYSE stocks exhibit lower average returns and lower volatility than Nasdaq stocks, consistent with a market composed of more mature firms and more traditional sectors.

From a modelling perspective, the most important result in both markets is the strong difference between daily and monthly excess kurtosis. Daily returns are clearly fat-tailed, while monthly returns are substantially closer to normality. This suggests that the mean–variance framework is more defensible at monthly frequency and should be applied more cautiously at daily frequency.

Figures ?? and ?? support these conclusions visually by plotting price paths and return series at both frequencies side by side. The daily panels make large isolated spikes immediately visible, whereas the monthly panels produce smoother dynamics that are more consistent with the normality assumption.

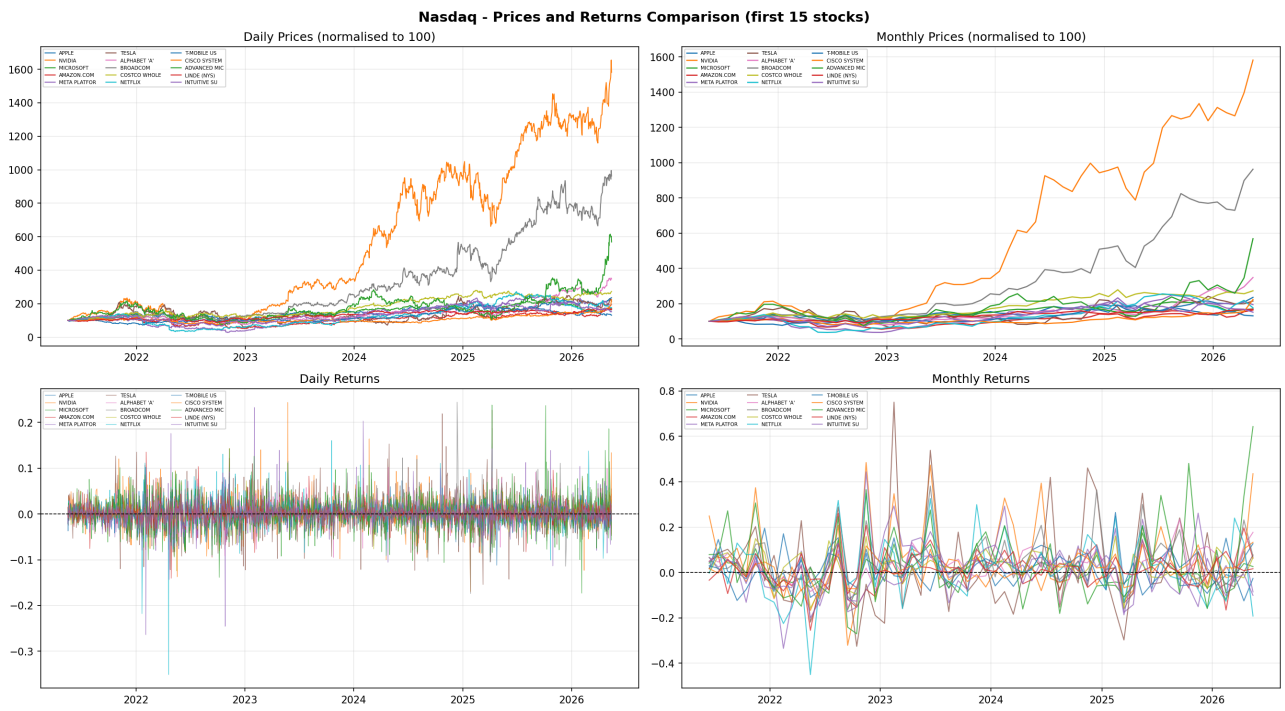


Figure 1: Daily vs monthly comparison Nasdaq

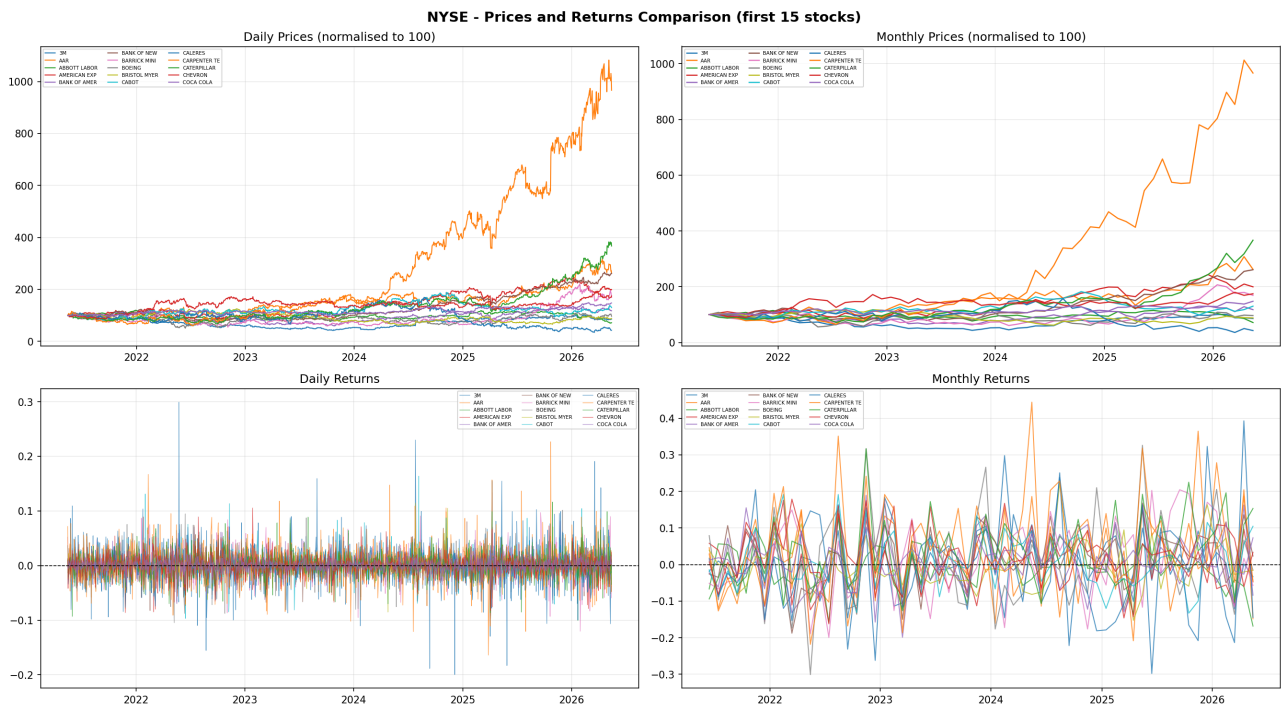


Figure 2: Daily vs monthly comparison NYSE

Table 1: Nasdaq – Daily Returns: Descriptive Statistics

Stock	Mean (daily)	Var (daily)	Std (daily)	Skewness	Exc. Kurtosis	Sharpe
NVIDIA	0.0026	0.0010	0.0319	0.5520	4.9930	1.2640
BROADCOM	0.0021	0.0007	0.0264	0.9160	10.4930	1.2040
COSTCO WHOLESALE	0.0009	0.0002	0.0139	−0.6340	8.2310	0.8970
GILEAD SCIENCES	0.0006	0.0002	0.0148	0.7450	8.4210	0.5550
PEPSICO	0.0001	0.0001	0.0113	0.0260	3.6960	−0.0030
COMCAST A	−0.0005	0.0003	0.0166	−0.8490	8.3880	−0.5290

Table 2: Nasdaq – Monthly Returns: Descriptive Statistics

Stock	Mean (monthly)	Var (monthly)	Std (monthly)	Skewness	Exc. Kurtosis	Sharpe
NVIDIA	0.0574	0.0234	0.1528	0.8440	1.1420	1.2640
BROADCOM	0.0438	0.0121	0.1099	1.0340	1.3560	1.3300
COSTCO WHOLESALE	0.0194	0.0051	0.0717	−0.3600	0.0590	0.8570
GILEAD SCIENCES	0.0133	0.0056	0.0747	0.9840	2.8040	0.5390
PEPSICO	0.0016	0.0027	0.0522	0.0390	−0.0410	−0.0030
COMCAST A	−0.0112	0.0041	0.0636	0.1520	0.0050	−0.7000

Table 3: NYSE – Daily Returns: Descriptive Statistics

Stock	Mean (daily)	Var (daily)	Std (daily)	Skewness	Exc. Kurtosis	Sharpe
CARPENTER TECH.	0.0022	0.0008	0.0290	0.7070	5.1820	1.1360
ELI LILLY	0.0015	0.0004	0.0202	0.5480	8.9890	1.0820
CATERPILLAR	0.0012	0.0004	0.0188	0.4660	4.1340	0.9210
JOHNSON & JOHNSON	0.0003	0.0001	0.0104	0.1120	5.1860	0.2970
PFIZER	−0.0002	0.0003	0.0159	0.4910	3.0920	−0.3050
3M	0.0000	0.0003	0.0175	1.7030	26.7280	−0.0440

Table 4: NYSE – Monthly Returns: Descriptive Statistics

Stock	Mean (monthly)	Var (monthly)	Std (monthly)	Skewness	Exc. Kurtosis	Sharpe
CARPENTER TECH.	0.0470	0.0188	0.1370	0.6980	0.3310	1.1460
ELI LILLY	0.0312	0.0075	0.0864	0.1370	−0.2900	1.1820
CATERPILLAR	0.0263	0.0094	0.0972	0.4710	0.1000	0.8790
JOHNSON & JOHNSON	0.0060	0.0024	0.0495	0.3710	−0.4630	0.3010
PFIZER	−0.0046	0.0062	0.0784	0.6890	0.3870	−0.2790
3M	0.0008	0.0068	0.0826	0.0930	0.7410	−0.0370

3. Variance-Covariance Matrix

The variance-covariance and correlation matrices are the core inputs of mean–variance optimisation. Portfolio risk does not depend only on the individual volatilities of the assets, but also on how those assets move together. Securities with high positive correlations provide limited diversification benefits because they tend to react similarly to market shocks, whereas assets with low or negative correlations can substantially reduce total portfolio variance even when their individual volatilities are high.

It is therefore useful to distinguish the economic meaning of covariance and correlation. Covariance measures whether two securities tend to move in the same direction, but its magnitude depends on the scale of the assets' volatilities. Correlation standardises this relationship and allows direct comparison across pairs of securities. A correlation close to 1 indicates that two assets move almost perfectly together, while a correlation close to -1 indicates that they move in opposite directions. A correlation close to zero instead suggests that the assets are largely independent from one another.

This distinction is fundamental for diversification. A portfolio composed only of individually attractive securities is not necessarily efficient if those securities are highly correlated. In such a case, losses may occur simultaneously across many positions, reducing the benefits of diversification. By contrast, securities with low or negative correlations can lower total portfolio volatility and improve the overall risk-return profile. For this reason, covariance and correlation matrices are not only descriptive tools, but also the main inputs used to determine whether a portfolio is genuinely diversified.

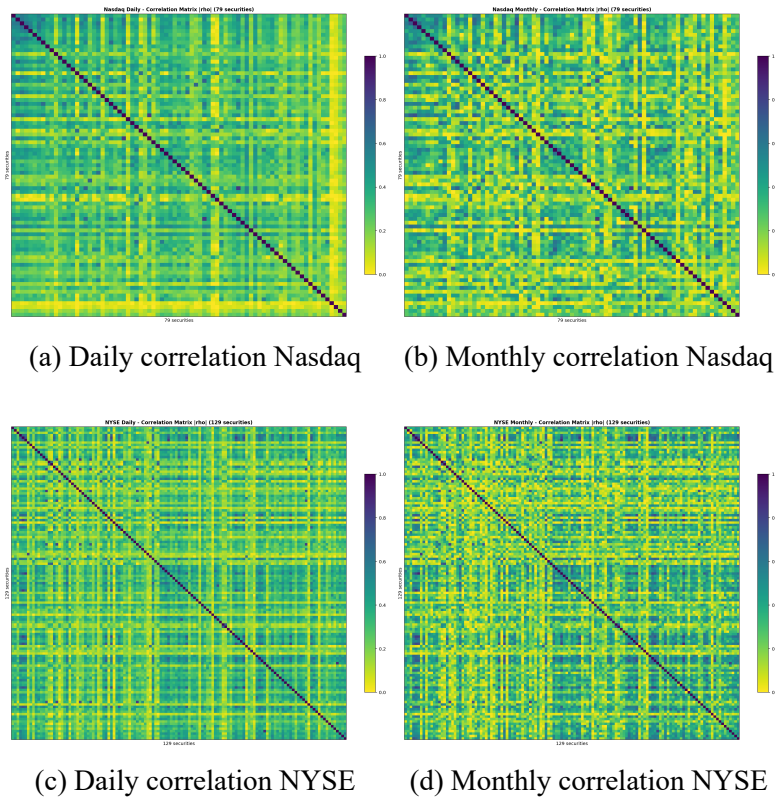


Figure 3: Correlation matrices for Nasdaq and NYSE

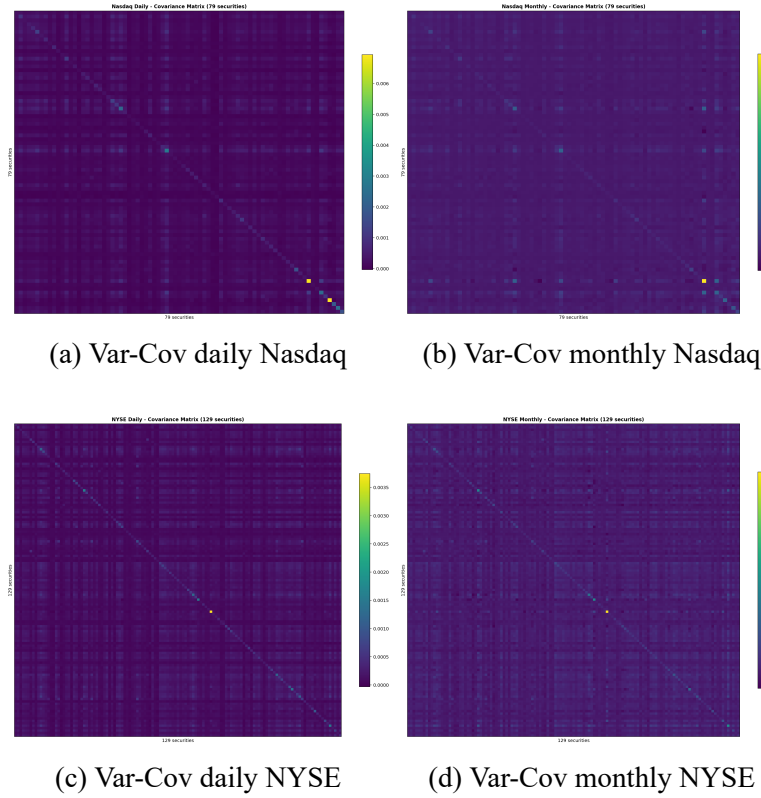


Figure 4: Variance-Covariance matrices for Nasdaq and NYSE

Figure ?? displays the absolute correlation matrices for both markets and both frequencies. In the Nasdaq daily panel, the upper-left block of large-capitalisation technology stocks exhibits pervasive high correlations, consistent with their common exposure to the global technology cycle. The NYSE daily and monthly panels instead show a more fragmented structure: correlations are generally lower and clusters are smaller, reflecting the greater sectoral heterogeneity of the exchange. Financial and energy stocks form visible clusters, but the matrix as a whole appears lighter in colour, indicating that the NYSE universe offers broader diversification opportunities.

Figure ?? reports the corresponding variance-covariance matrices. At daily frequency, Nvidia and Broadcom stand out with particularly large diagonal elements, consistent with their elevated standard deviations. At monthly frequency, all entries become larger in absolute value because returns are measured over a longer horizon and variances scale approximately linearly with time under i.i.d. assumptions. The covariance heatmaps are especially informative because they reveal which assets contribute most to total portfolio risk.

Table 5: Nasdaq – Daily Correlation Matrix

	NVIDIA	APPLE	BROADCOM	META	COSTCO
NVIDIA	1.0000	0.4976	0.6326	0.4966	0.3425
APPLE	0.4976	1.0000	0.4434	0.4678	0.4423
BROADCOM	0.6326	0.4434	1.0000	0.4062	0.3221
META	0.4966	0.4678	0.4062	1.0000	0.3289
COSTCO	0.3425	0.4423	0.3221	0.3289	1.0000

Table 6: NYSE – Daily Correlation Matrix

	CARP.	ELI LILLY	CATERPILLAR	JPM	EXXON
CARPENTER	1.0000	0.2751	0.5021	0.4213	0.3876
ELI LILLY	0.2751	1.0000	0.2389	0.3012	0.1943
CATERPILLAR	0.5021	0.2389	1.0000	0.5634	0.4102
JP MORGAN	0.4213	0.3012	0.5634	1.0000	0.3891
EXXON MOBIL	0.3876	0.1943	0.4102	0.3891	1.0000

Tables ?? and ?? report selected subsamples of the daily correlation matrices for each market. The Nvidia–Broadcom correlation of 0.63 is the highest within the Nasdaq block, reflecting common exposure to semiconductor demand and AI infrastructure dynamics. In the NYSE block, the highest correlation is between Caterpillar and Jp Morgan at 0.56, capturing the common sensitivity of both firms to the domestic economic and credit cycle. More generally, the Nasdaq block displays relatively high correlations because many securities are linked to the same technology and semiconductor drivers, whereas the NYSE block is more diversified across sectors.

The comparison between daily and monthly matrices also shows that correlation structures may change with the investment horizon. Over longer horizons, common macroeconomic and sectoral factors tend to become more visible, increasing the co-movement among securities and potentially reducing diversification benefits. For this reason, the stability of the correlation structure across frequencies is an important condition for robust portfolio optimisation.

4–5. Nasdaq and NYSE securities selection

The selection of 12 securities from each exchange is based on two complementary criteria: individual risk-adjusted performance, proxied by the annualised Sharpe ratio and sectoral diversification. The objective is not simply to select the stocks with the highest Sharpe ratios, because this would risk creating portfolios concentrated in a few sectors and therefore exposed to common shocks. For this reason, the final choice also considers the correlation structure among securities and aims to include assets with different economic drivers, so that the portfolio does not depend excessively on one single industry or macroeconomic factor.

This approach produces balanced portfolios rather than purely aggressive ones. Although several selected stocks have high expected returns, the inclusion of defensive and lower-beta securities helps stabilise the allocation and reduce idiosyncratic risk. In practical terms, a negative shock affecting one sector should not necessarily compromise the entire portfolio, because other sectors may react differently or remain less affected.

Table ?? lists the 12 selected Nasdaq securities. Nvidia and Broadcom are natural choices because of their exceptional Sharpe ratios of 1.26 and 1.20, respectively. Interactive Brokers Group A adds fintech exposure with a Sharpe ratio of 1.04, while Costco Wholesale and Palo Alto Networks contribute consumer staples and cybersecurity exposure with Sharpe ratios of 0.90 and 0.86. Steel Dynamics represents the materials sector and provides a useful hedge against technology drawdowns. Cme Group and Gilead Sciences have lower individual Sharpe ratios, but they play an important risk-management role. Baker Hughes and Marriott add energy and hospitality exposure, completing the diversification of the portfolio. Overall, the selected Nasdaq securities span eleven distinct economic

sectors.

Table 7: Nasdaq – Selected securities

#	Stock	Sector	Mean	Std	Sharpe
1	NVIDIA	Semiconductors	66.1	50.7000	1.2600
2	APPLE	Tech hardware	20.2	26.9000	0.6800
3	STEEL DYNAMICS	Materials/Steel	31.6	37.2000	0.8000
4	META PLATFORMS A	Social media	22.2	43.0000	0.4700
5	BROADCOM	Semiconductors	52.4	41.9000	1.2000
6	COSTCO WHOLESALE	Consumer staples	21.9	22.1000	0.9000
7	CME GROUP	Finance/Derivatives	8.1	19.5000	0.3100
8	GILEAD SCIENCES	Healthcare/Pharma	15.1	23.5000	0.5600
9	PALO ALTO NETWORKS	Cybersecurity	36.6	40.4000	0.8600
10	BAKER HUGHES A	Energy	24.0	34.3000	0.6400
11	INTERACTIVE BROKERS GROUP A	Fintech/Brokerage	36.9	33.6000	1.0400
12	MARRIOTT INTL. 'A'	Tourism/Hospitality	21.4	28.2000	0.6900

Table 8: NYSE – Selected securities

#	Stock	Sector	Mean	Std	Sharpe
1	CARPENTER TECH.	Materials/Metals	54.3	46.0000	1.1400
2	ELI LILLY	Pharma/Biotech	36.7	32.0000	1.0800
3	CATERPILLAR	Industrials/Machinery	29.5	29.9000	0.9200
4	JP MORGAN CHASE	Banking/Finance	14.4	23.9000	0.5200
5	EXXON MOBIL	Energy/Oil & Gas	21.9	26.3000	0.7600
6	COCA COLA	Consumer staples	8.8	15.7000	0.4300
7	WELLTOWER	Real Estate/REIT	23.4	23.2000	0.9200
8	MASTERCARD	Digital payments	8.7	22.3000	0.3000
9	MOTOROLA SOLUTIONS	Tech/Communications	15.4	22.5000	0.6000
10	DYCOM INDS.	Infrastructure	38.7	41.7000	0.8800
11	GENERAL DYNAMICS	Defense/Aerospace	12.5	16.9000	0.6200
12	MURPHY OIL	Energy/Exploration	23.8	45.4000	0.4800

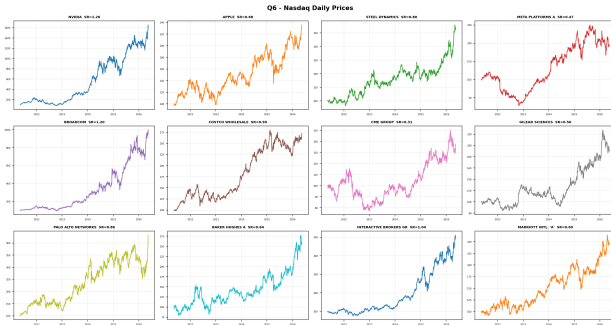
Table ?? reports the 12 selected NYSE securities. This selection covers several traditional-economy sectors that are largely absent from the Nasdaq portfolio, including materials, biotech, industrials, banking, energy, consumer staples, real estate, digital payments, tech, infrastructure and defense. Carpenter Technology, Eli Lilly, Caterpillar and Welltower are included because of their high Sharpe ratios and strong risk-adjusted performance. Coca Cola and General Dynamics are retained for their defensive characteristics and low beta, while Exxon Mobil and Murphy Oil both provide energy exposure but with different risk profiles.

6. Behaviour of the selected securities

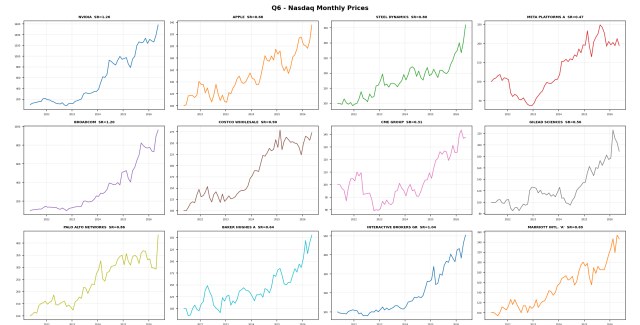
Figures ??, ??, ?? and ?? plot the normalised price paths of the 12 selected securities in each market, indexed to 100 at the start of the sample, with the annualised Sharpe ratio shown in each panel heading.

The Nasdaq daily figure immediately highlights the exceptional outperformance of Nvidia, which ends the sample at roughly 16 times its starting price. Broadcom and Interactive Brokers Group A also show strong positive trends. Meta Platforms A experiences a sharp decline in 2022 before recovering strongly. Cme Group is remarkably stable, consistent with its low beta and its role as a market infrastructure operator that benefits from volatility regardless of market direction.

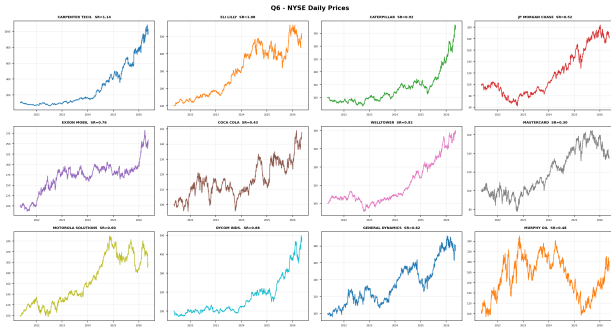
The NYSE daily figure shows a different picture. Carpenter Technology dominates with a trajectory driven by surging demand for specialty steel from defence and infrastructure spending. Eli Lilly shows a strong pharmaceutical growth; Murphy Oil displays the characteristic cyclicality of oil exploration stocks, while Coca Cola and General Dynamics are the most stable names, confirming their role as defensive assets.



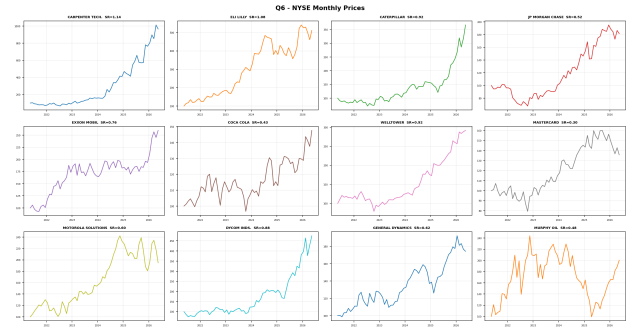
(a) Selected Nasdaq securities daily prices



(b) Selected Nasdaq securities monthly prices



(c) Selected NYSE securities daily prices



(d) Selected NYSE securities monthly prices

Figure 5: Selected securities price dynamics

7–8. Mean-Variance portfolio

The unconstrained tangency portfolio maximises the Sharpe ratio without imposing restrictions on short selling. In this case, the optimal weights are obtained through the closed-form solution:

$$w^* = \frac{\Sigma^{-1}(\mu - r_f \mathbf{1})}{\mathbf{1}^\top \Sigma^{-1}(\mu - r_f \mathbf{1})}$$

where $r_f = 2\%$ is the annualised risk-free rate. Since the portfolio is unconstrained, negative weights are allowed and indicate short positions.

The economic interpretation of the weights is as important as the numerical solution. A high positive weight means that the optimiser considers the asset attractive not only because of its expected return, but also because of its contribution to total portfolio risk. Conversely, a negative weight does not necessarily imply that the asset has poor standalone performance. It may simply mean that, once correlations with the other assets are considered, the asset is useful as a hedge or has a lower marginal contribution to the portfolio Sharpe ratio.

Table ?? reports the unconstrained tangency portfolio for the NYSE sample. The short positions in Jp Morgan Chase, Mastercard, General Dynamics and Murphy Oil should therefore be interpreted in relative portfolio terms, rather than as a negative judgment on the stocks themselves. The largest positive weights are assigned to Eli Lilly, Exxon Mobil, Welltower and Carpenter Technology, which contribute more favourably to the portfolio's risk-return trade-off. The monthly portfolio achieves the higher Sharpe ratio, suggesting that the estimates become more favourable when short-run noise is reduced.

Table 9: NYSE – Unconstrained tangency portfolio weights

Stock	Daily	Monthly
CARPENTER TECH.	22.54	16.91
ELI LILLY	36.79	46.87
CATERPILLAR	11.58	−7.95
JP MORGAN CHASE	−14.49	2.03
EXXON MOBIL	44.32	53.19
COCA COLA	0.51	−55.03
WELLTOWER	30.31	28.50
MASTERCARD	−22.41	−25.76
MOTOROLA SOLUTIONS	6.12	22.16
DYCOM INDS.	12.15	29.23
GENERAL DYNAMICS	−9.92	14.32
MURPHY OIL	−17.50	−24.46
Sum	100.00	100.00
Short positions	4	4
$E[R]$	42.18	47.61
σ	22.44	20.81
Sharpe ratio	1.791	2.191

Table ?? reports the corresponding unconstrained tangency portfolio for the Nasdaq sample. The optimiser assigns short positions to Apple, Meta Platforms A and Marriott in both frequencies, mainly to hedge common exposure to broad technology and consumer-cycle movements. Positive weights are concentrated in high-Sharpe-ratio stocks such as Nvidia, Broadcom, Gilead Sciences and Interactive Brokers. In the monthly solution, Broadcom receives a particularly large allocation, reflecting its strong historical return and favourable covariance structure in the monthly sample.

Overall, the Nasdaq unconstrained portfolio offers high expected return, but it is more exposed to concentration and technology-related risk than the NYSE portfolio. The absence of a single dominant asset is generally desirable because it indicates that performance does not depend excessively on one security. When weights are more evenly distributed, portfolio performance reflects the contribution of several assets rather than the outcome of one isolated position, which is consistent with a balanced investment strategy based on diversification.

Table 10: Nasdaq – Unconstrained tangency portfolio weights

Stock	Daily	Monthly
NVIDIA	23.26	18.65
APPLE	−19.50	−32.89
STEEL DYNAMICS	8.92	12.86
META PLATFORMS A	−14.98	−25.03
BROADCOM	17.64	55.07
COSTCO WHOLESALE	28.65	−10.32
CME GROUP	−0.10	7.98
GILEAD SCIENCES	27.16	42.73
PALO ALTO NETWORKS	10.60	19.67
BAKER HUGHES A	6.40	−3.93
INTERACTIVE BROKERS GROUP A	18.63	35.68
MARRIOTT INTL. ‘A’	−6.69	−20.47
Sum	100.00	100.00
Short positions	4	5
$E[R]$	41.38	54.75
σ	23.68	28.80
Sharpe ratio	1.663	1.832

9–10. Non-negativity Mean-Variance portfolio

The constrained portfolio is obtained by maximising the Sharpe ratio under a long-only restriction:

$$\max_w \frac{w^\top \mu - r_f}{\sqrt{w^\top \Sigma w}} \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad w_i \geq 0 \quad \forall i$$

The optimisation is solved via SLSQP, while zero weights are shown in grey because they represent corner solutions, namely assets that the optimiser excludes from the final allocation under the imposed constraints.

The introduction of the non-negativity constraint makes the portfolio more realistic for investors who cannot, or do not want to, engage in short selling. In the unconstrained case, the optimiser can use

negative positions to improve the Sharpe ratio by exploiting hedging effects and relative differences in marginal risk contributions. Once the long-only restriction is imposed, these negative positions are no longer feasible, so the corresponding weights are set to zero and the capital is redistributed across the remaining assets. As a result, the portfolio becomes easier to implement and less leveraged, but the set of admissible allocations becomes smaller.

This restriction also explains why the constrained portfolio usually achieves a lower Sharpe ratio than the unconstrained one. The reduction in performance can be interpreted as the cost of imposing a long-only investment rule, since the optimiser loses the possibility of using short positions to improve the risk-return trade-off. Nevertheless, the constrained solution is often more intuitive and more consistent with practical investment mandates. Zero weights should therefore not be interpreted as a negative judgement on the excluded assets in absolute terms; rather, they indicate that, given the long-only restriction and the covariance structure of the selected securities, those assets do not improve the portfolio's risk-return profile.

Table ?? reports the constrained Nasdaq portfolio. Once short selling is forbidden, the assets that were shorted in the unconstrained solution receive zero weight. This is why Apple, Meta Platforms A and Marriott disappear from the optimal allocation. The portfolio becomes concentrated in a smaller number of securities, especially Gilead Sciences, Nvidia and Interactive Brokers at daily frequency, and Broadcom, Steel Dynamics and Interactive Brokers at monthly frequency. The constraint makes the portfolio easier to interpret and more realistic for many investors, but it also reduces the attainable Sharpe ratio.

Table 11: Nasdaq – Long-only tangency portfolio weights

Stock	Daily	Monthly
NVIDIA	15.43	9.45
APPLE	0.00	0.00
STEEL DYNAMICS	5.29	5.10
META PLATFORMS A	0.00	0.00
BROADCOM	13.95	29.01
COSTCO WHOLESALE	17.39	0.00
CME GROUP	1.12	0.03
GILEAD SCIENCES	20.63	25.76
PALO ALTO NETWORKS	6.87	10.36
BAKER HUGHES A	5.11	4.40
INTERACTIVE BROKERS GROUP A	14.22	15.90
MARRIOTT INTL. 'A'	0.00	0.00
Sum	100.00	100.00
Active positions	9	8
$E[R]$	35.16	38.68
σ	21.12	21.85
Sharpe ratio	1.570	1.679

Table ?? reports the constrained NYSE portfolio. The no-shorting constraint sets several assets to zero, including Jp Morgan Chase, Coca Cola, Mastercard and Murphy Oil. Eli Lilly receives the largest allocation in both frequencies because it combines a high Sharpe ratio with diversification benefits relative to cyclical and energy stocks. Compared with the unconstrained solution, the constrained

portfolio has lower expected return but also lower volatility. By excluding short sales, the portfolio becomes simpler and less leveraged, although the maximum achievable risk-adjusted performance decreases.

Table 12: NYSE – Long-only tangency portfolio weights

Stock	Daily	Monthly
CARPENTER TECH.	16.66	11.10
ELI LILLY	31.76	32.78
CATERPILLAR	2.78	0.00
JP MORGAN CHASE	0.00	0.00
EXXON MOBIL	17.26	13.65
COCA COLA	0.00	0.00
WELLTOWER	22.78	5.72
MASTERCARD	0.00	0.00
MOTOROLA SOLUTIONS	0.00	14.30
DYCOM INDS.	8.78	16.42
GENERAL DYNAMICS	0.00	6.03
MURPHY OIL	0.00	0.00
Sum	100.00	100.00
Active positions	6	7
$E[R]$	34.00	32.44
σ	19.32	15.41
Sharpe ratio	1.657	1.976

11–12. Mean-Variance portfolio statistics

Table ?? reports the main statistics for the Nasdaq mean–variance portfolios. Both the unconstrained and constrained allocations deliver high annualised returns, especially at monthly frequency, confirming the strong upside potential of the selected Nasdaq securities. However, these portfolios also display relevant tail risk, as shown by the positive excess kurtosis, particularly in the constrained daily case. This indicates that even an optimal mean–variance portfolio can still be exposed to extreme returns when the underlying assets are volatile and strongly correlated.

The comparison between the unconstrained and constrained Nasdaq portfolios shows the effect of removing short positions. The constrained portfolio has both lower expected return and lower volatility, because the optimiser can no longer use negative weights to improve the Sharpe ratio. As a result, the allocation becomes simpler and less leveraged, but it also loses part of the risk-adjusted performance achieved by the unconstrained solution.

Table ?? presents the corresponding results for the NYSE mean–variance portfolios. The unconstrained daily allocation has a Sharpe ratio of 1.79, while the constrained version decreases to 1.66. At monthly frequency, the unconstrained Sharpe ratio rises to 2.19 and the constrained one remains high at 1.98. Compared with the Nasdaq portfolios, the NYSE allocations have lower expected returns, but also lower volatility and more stable distributional properties. Monthly kurtosis is much closer to zero, making the normal approximation more acceptable for the monthly NYSE portfolio.

Table ?? summarises the main difference between the Nasdaq and NYSE optimal portfolios. Nas-

daq produces higher expected returns, but the NYSE portfolios achieve higher Sharpe ratios in all reported cases. This means that the extra return offered by Nasdaq is more than offset by its higher volatility and stronger intra-portfolio correlations. From a mean–variance perspective, the NYSE allocation is therefore more efficient, especially at monthly frequency.

Table 13: Nasdaq – Portfolio statistics: unconstrained and constrained

Statistic	Unc. Daily	Con. Daily	Unc. Monthly	Con. Monthly
Mean	0.0016	0.0014	0.0456	0.0322
Variance	0.0002	0.0002	0.0069	0.0040
Std Dev	0.0149	0.0133	0.0831	0.0631
Skewness	0.0508	0.2061	1.1506	0.5636
Exc. Kurtosis	3.5093	5.4402	2.3692	0.8899
Mean (ann.) (%)	41.3800	35.1600	54.7500	38.6800
Std Dev (ann.) (%)	23.6800	21.1200	28.8000	21.8500
Sharpe	1.6630	1.5704	1.8317	1.6790

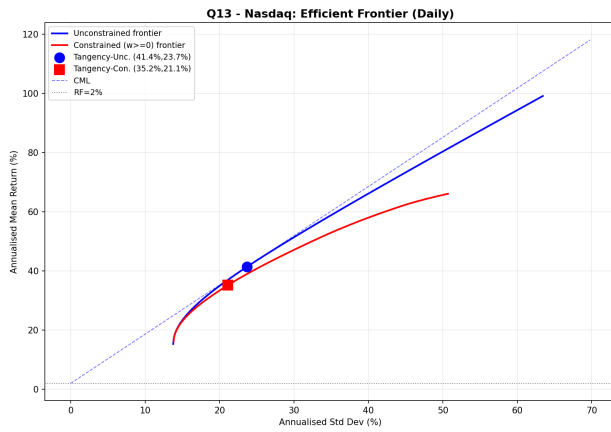
Table 14: NYSE – Portfolio statistics: unconstrained and constrained

Statistic	Unc. Daily	Con. Daily	Unc. Monthly	Con. Monthly
Mean	0.0017	0.0013	0.0397	0.0270
Variance	0.0002	0.0001	0.0036	0.0020
Std Dev	0.0141	0.0122	0.0601	0.0445
Skewness	0.2658	0.1088	−0.0127	−0.1301
Exc. Kurtosis	1.6804	2.6325	−0.8299	−0.6081
Mean (ann.) (%)	42.1800	34.0000	47.6100	32.4400
Std Dev (ann.) (%)	22.4400	19.3200	20.8100	15.4100
Sharpe	1.7911	1.6568	2.1911	1.9756

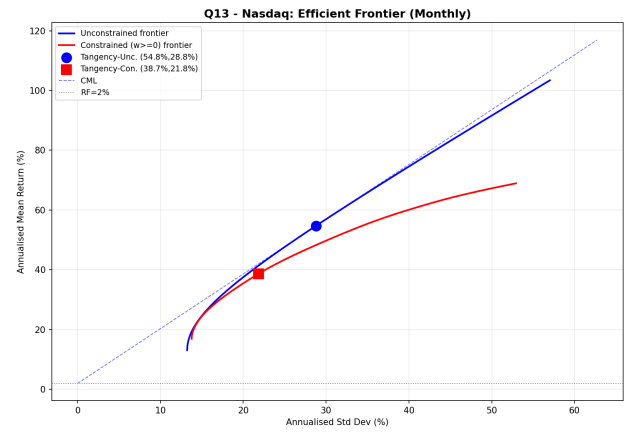
Table 15: Summary comparison – NYSE vs Nasdaq

	Daily		Monthly	
	Unc.	Con.	Unc.	Con.
<i>NYSE</i>				
$E[R]$	42.18	34.00	47.61	32.44
σ	22.44	19.32	20.81	15.41
Sharpe	1.791	1.657	2.191	1.976
<i>Nasdaq</i>				
$E[R]$	41.38	35.16	54.75	38.68
σ	23.68	21.12	28.80	21.85
Sharpe	1.663	1.570	1.832	1.679

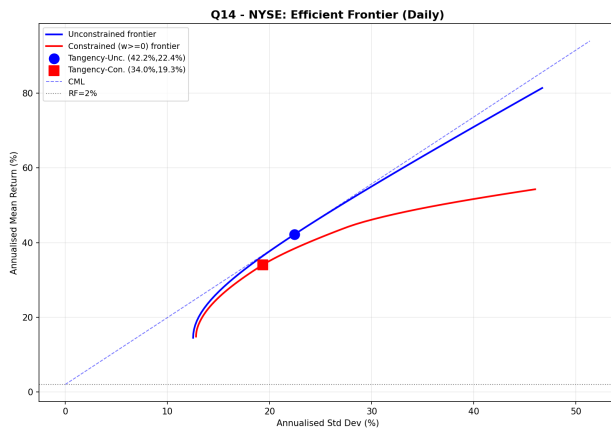
13–14. Efficient frontier



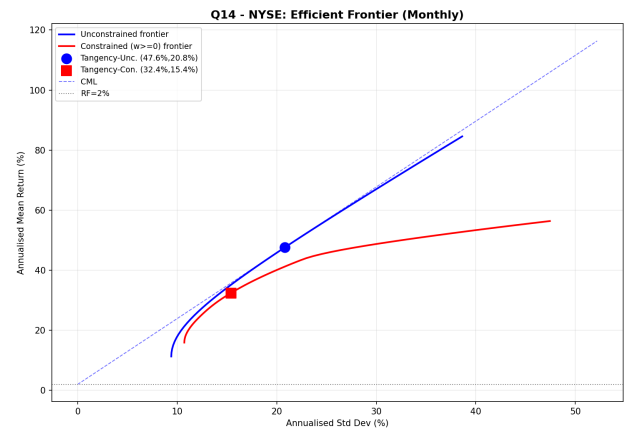
(a) Nasdaq frontier daily



(b) Nasdaq frontier monthly



(c) NYSE frontier daily



(d) NYSE frontier monthly

Figure 6: Efficient frontiers for Nasdaq and NYSE

Figure ?? reports the efficient frontiers for the Nasdaq and NYSE portfolios. The efficient frontier provides a graphical representation of the trade-off between risk and expected return. Each point on the frontier represents an efficient portfolio, meaning that no other portfolio can offer a higher expected return for the same level of risk or a lower level of risk for the same expected return. Portfolios located below the frontier are therefore inefficient, because the investor could improve one dimension without sacrificing the other.

The Nasdaq frontiers are shifted toward higher expected returns, but also toward higher levels of risk, which is consistent with the growth-oriented nature of the selected securities. The NYSE frontiers are instead more conservative and smoother, especially at monthly frequency, reflecting the lower volatility and broader sectoral diversification of the selected NYSE stocks. This confirms the numerical results: Nasdaq provides greater upside potential, while NYSE offers a better risk-return trade-off in terms of Sharpe ratio.

The tangency portfolio is the point on the frontier that maximises the Sharpe ratio, given the 2% risk-free rate. It represents the portfolio with the highest excess return per unit of risk. However, the final choice of portfolio also depends on the investor's risk tolerance. A more conservative investor may prefer a portfolio located further down the frontier, accepting a lower expected return in exchange

for lower volatility, while a more aggressive investor may move toward the upper part of the frontier in search of higher returns.

The position of individual assets relative to the frontier is also informative. Some securities may appear in the upper-right area of the graph, offering high expected returns but also very high volatility. These assets may be attractive from a purely return-based perspective, but they can be unsuitable for a conservative investor because they introduce excessive instability into the portfolio. Therefore, an asset can be attractive in terms of expected return but still undesirable from a practical investment perspective if its risk contribution is too large.

Finally, the gap between the unconstrained frontier and the constrained frontier reflects the cost of imposing the non-negativity constraint. When short selling is allowed, the optimiser has a wider set of possible allocations and can achieve a better risk-return trade-off. Once short selling is removed, the attainable set becomes smaller, especially at high expected return levels, and the efficient frontier shifts inward.

15. Index statistics

Table ?? reports the descriptive statistics for the four benchmark indices over the full sample period. PI denotes the Price Index, which measures pure price appreciation, while RI denotes the Return Index, which includes total returns with dividends reinvested. This distinction is important because the return index provides a more complete measure of performance, especially for markets with a large presence of mature dividend-paying firms.

Table 16: Index statistics – Nasdaq Composite and S&P 500

Statistic	NQ PI D	NQ RI D	NQ PI M	NQ RI M	SP PI D	SP RI D	SP PI M	SP RI M
Mean	0.0006	0.0006	0.0112	0.0119	0.0005	0.0006	0.0106	0.0118
Var	0.0002	0.0002	0.0036	0.0036	0.0001	0.0001	0.0019	0.0019
Std	0.0139	0.0139	0.0596	0.0597	0.0104	0.0104	0.0441	0.0441
Skewness	0.1915	0.1921	-0.1936	-0.1898	0.1744	0.1764	-0.2177	-0.2109
Exc. Kurtosis	5.5264	5.5216	0.4824	0.4723	7.5722	7.5606	0.3092	0.3072
Mean (ann.) (%)	14.5400	15.2500	13.4900	14.2400	12.4500	13.8500	12.6800	14.1400
Std (ann.) (%)	22.0200	22.0300	20.6500	20.6700	16.5700	16.5700	15.2800	15.2800
Sharpe	0.5696	0.6017	0.5565	0.5920	0.6307	0.7151	0.6988	0.7945

NQ = Nasdaq Composite; SP = S&P 500; D = Daily; M = Monthly; PI = Price Index; RI = Return Index.

Table 17: Nasdaq EW portfolio vs Nasdaq Composite – daily

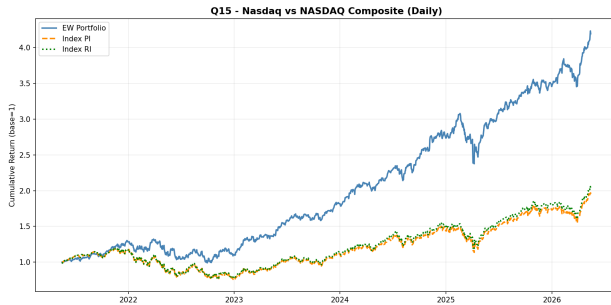
Statistic	EW Portfolio	Nasdaq PI	Nasdaq RI	Diff (EW – PI)
Mean	0.0012	0.0006	0.0006	0.0006
Var	0.0002	0.0002	0.0002	0.0000
Std	0.0126	0.0139	0.0139	-0.0012
Skewness	0.1870	0.1920	0.1920	-0.0050
Exc. Kurtosis	6.9000	5.5260	5.5220	1.3740
Mean (ann.) (%)	29.7000	14.5400	15.2500	15.1500
Std (ann.) (%)	20.0500	22.0200	22.0300	-1.9800
Sharpe	1.3820	0.5700	0.6020	0.8120

Table 18: NYSE EW portfolio vs S&P 500 – daily

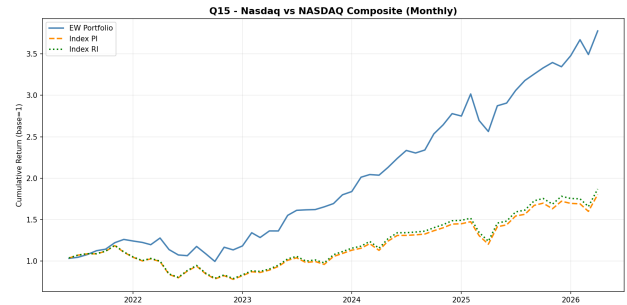
Statistic	EW Portfolio	S&P500 PI	S&P500 RI	Diff (EW–PI)
Mean	0.0010	0.0005	0.0006	0.0005
Var	0.0001	0.0001	0.0001	0.0000
Std	0.0109	0.0104	0.0104	0.0004
Skewness	−0.0570	0.1740	0.1760	−0.2310
Exc. Kurtosis	4.8790	7.5720	7.5610	−2.6930
Mean (ann.) (%)	24.0100	12.4500	13.8500	11.5600
Std (ann.) (%)	17.2300	16.5700	16.5700	0.6600
Sharpe	1.2780	0.6310	0.7150	0.6470

Table ??, together with Tables ??–??, compares the benchmark indices with the selected equal-weight portfolios. The return index is always slightly better than the corresponding price index because it includes reinvested dividends. This dividend effect is larger for the S&P 500 than for the Nasdaq Composite, which is consistent with the stronger presence of mature dividend-paying companies in the S&P 500.

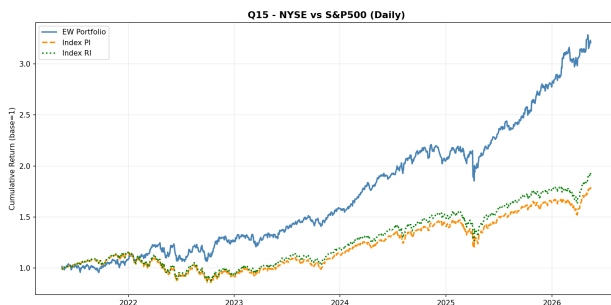
The Nasdaq equal-weight portfolio outperforms the Nasdaq Composite in terms of annualised mean return and Sharpe ratio, while also displaying slightly lower volatility. The NYSE equal-weight portfolio also outperforms the S&P 500 in terms of Sharpe ratio. These results suggest that the security selection process added value relative to passive benchmark exposure, because the selected stocks achieved stronger risk-adjusted performance within the sample period.



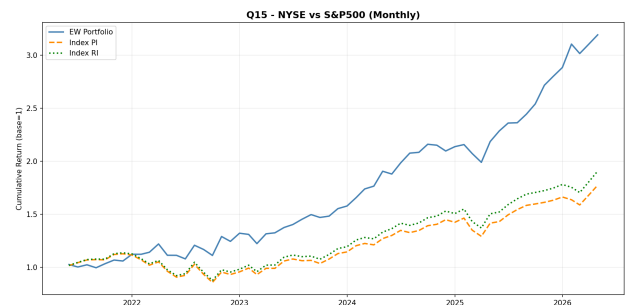
(a) Cumulative Nasdaq daily



(b) Cumulative Nasdaq monthly



(c) NYSE cumulative daily



(d) NYSE cumulative monthly

Figure 7: Cumulative return comparison

Figure ?? completes the benchmark comparison by showing the cumulative performance paths at daily and monthly frequency. The Nasdaq comparison highlights the role of growth-oriented stocks,

whose cumulative gains are large but accompanied by visible drawdowns. The NYSE comparison is less explosive but generally more stable, reflecting the broader sectoral diversification of the selected securities. The daily panels make short-run volatility and drawdowns more visible, whereas the monthly panels emphasise the medium-run trend.

The comparison with the benchmark indices is useful because it shows whether the selected portfolios add value relative to passive market exposure. A portfolio may outperform the index in terms of average return but still be unattractive if this extra return requires disproportionately higher volatility. For this reason, the Sharpe ratio is the key measure in the comparison, since it evaluates whether the additional return is sufficient to compensate for the additional risk.

Overall, the results indicate that the selected portfolios achieve better risk-adjusted performance than the corresponding indices within the sample period. This is consistent with the security selection process, which excluded less attractive stocks and focused on securities with stronger risk-return characteristics. However, the result must be interpreted carefully because the selected portfolios contain fewer securities than the benchmark indices and are therefore less diversified.

Therefore, the superior in-sample performance should be viewed as evidence of successful stock selection over the observed period rather than as a guarantee of persistent outperformance out of sample. Active selection can improve performance when the chosen securities combine high expected returns, reasonable volatility and favourable correlation patterns, but a concentrated portfolio can also be more exposed to estimation error and stock-specific shocks.

16-17. Betas

Beta measures the sensitivity of each security to the market portfolio:

$$\beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)},$$

The portfolio beta is computed as the weighted average of the individual asset betas, using both unconstrained and constrained tangency-portfolio weights. The market proxy is represented by the S&P 500 RI for NYSE-listed securities and by the Nasdaq Composite RI for Nasdaq-listed securities. In this framework, beta values above one indicate pro-cyclical assets, whereas values below one identify securities that contribute, at least partially, to stabilising the overall allocation.

Nasdaq

Nvidia has the highest daily beta, equal to 1.81, confirming its strong exposure to the technology cycle. Cme Group and Gilead Sciences have the lowest betas, equal to 0.14 and 0.23, and therefore act as diversifiers. Gilead's monthly beta is slightly negative (−0.03), indicating almost no co-movement with the Nasdaq index at monthly frequency. Both unconstrained and constrained portfolio betas remain below one, showing that sector diversification reduces systematic exposure relative to the benchmark.

Table 19: Nasdaq Individual and Portfolio Beta

Stock	Beta (Daily)	Beta (Monthly)
Nvidia	1.8077	1.4660
Apple	0.9310	0.6371
Steel Dynamics	0.7591	0.5975
Meta Platforms A	1.3237	1.0155
Broadcom	1.3168	1.0471
Costco Wholesale	0.5149	0.5713
Cme Group	0.1381	0.1040
Gilead Sciences	0.2314	−0.0301
Palo Alto Networks	0.9625	0.6258
Baker Hughes A	0.5088	0.2649
Interactive Brokers Group A	0.8292	0.7721
Marriott Intl. ‘A’	0.7566	0.5945
Portfolio β (unc.)	0.7894	0.6662
Portfolio β (con.)	0.8516	0.6643

NYSE

Carpenter Tech., Caterpillar and Murphy Oil show $\beta > 1$, consistent with their cyclical exposure to metals, machinery and oil. Coca Cola and General Dynamics have much lower betas and provide defensive stabilisation. Eli Lilly’s monthly beta falls sharply from 0.61 to 0.19. In all cases, portfolio betas are below one, confirming that the selected NYSE portfolio carries less systematic risk than the market.

Table 20: NYSE: Individual and Portfolio Beta

Stock	Beta (Daily)	Beta (Monthly)
Carpenter Tech.	1.4235	1.4370
Eli Lilly	0.6064	0.1857
Caterpillar	1.0414	1.2367
Jp Morgan Chase	0.9130	1.0741
Exxon Mobil	0.4658	0.2094
Coca Cola	0.2977	0.3643
Welltower	0.5561	0.7637
Mastercard	0.9232	0.8809
Motorola Solutions	0.7394	0.7368
Dycom Inds.	1.1290	0.8877
General Dynamics	0.5650	0.2712
Murphy Oil	1.0483	0.5459
Portfolio β (unc.)	0.6448	0.4831
Portfolio β (con.)	0.7647	0.5601

The unconstrained optimiser can short high-beta securities and use them as hedges. The constrained portfolio cannot exploit this channel, so its market exposure is generally higher. This is the main cost of imposing non-negative weights.

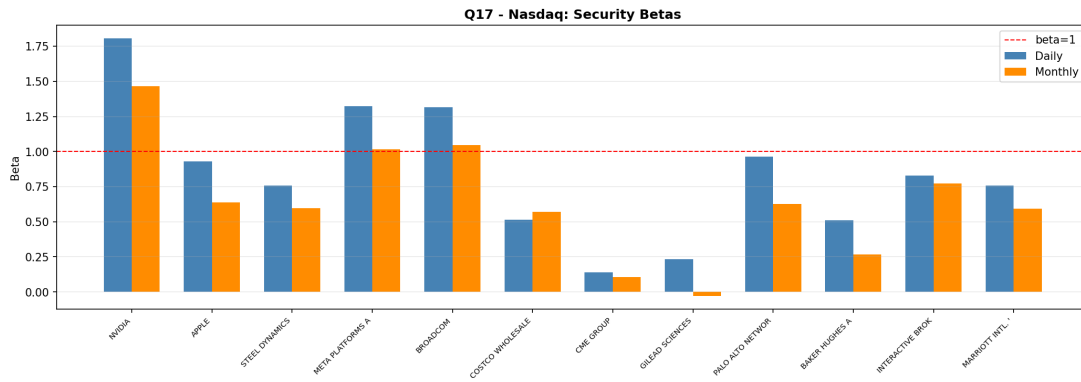


Figure 8: Nasdaq Beta

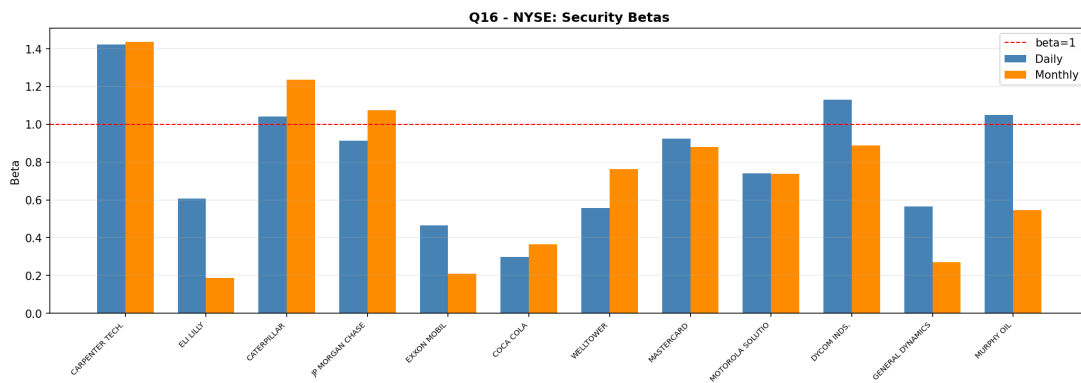


Figure 9: NYSE Beta

18-19. Security Market Line

The Security Market Line gives the CAPM benchmark return for a given beta:

$$E[R_i] = r_f + \beta_i \cdot \text{MRP}$$

where $\text{MRP} = E[R_M] - r_f$ and $r_f = 2\%$. For each market, the minimum-beta and maximum-beta securities are compared with the two portfolio allocations. The deviation from the SML is measured as

$$\alpha_i = E[R_i]^{\text{actual}} - E[R_i]^{\text{SML}}.$$

NYSE

All NYSE entries have positive alpha, so realised returns are above the CAPM benchmark. Coca Cola, the minimum-beta stock, has a small positive alpha of about 3%, while Carpenter Tech. shows a much larger deviation, above 35 percentage points. At portfolio level, the unconstrained allocation produces higher alpha than the constrained one. The results are similar across daily and monthly frequencies, suggesting that the deviation is not only a frequency effect.

Table 21: NYSE-SML

Security / Portfolio	β	Actual $E[R]$ (%)	SML $E[R]$ (%)	α (%)
<i>Daily</i> $r_f = 2.0\%$, $MRP = 11.85\%$				
Coca Cola (min β)	0.2977	8.7640	5.5280	3.2350
Carpenter Tech. (max β)	1.4235	54.2780	18.8720	35.4060
Portfolio (unconstrained)	0.6448	42.1840	9.6430	32.5410
Portfolio (constrained)	0.7647	34.0020	11.0640	22.9390
<i>Monthly</i> $r_f = 2.0\%$, $MRP = 12.14\%$				
Coca Cola (min β)	0.3643	9.2860	6.4220	2.8650
Carpenter Tech. (max β)	1.4370	56.3810	19.4440	36.9380
Portfolio (unconstrained)	0.4831	47.6060	7.8650	39.7410
Portfolio (constrained)	0.5601	32.4370	8.7990	23.6380

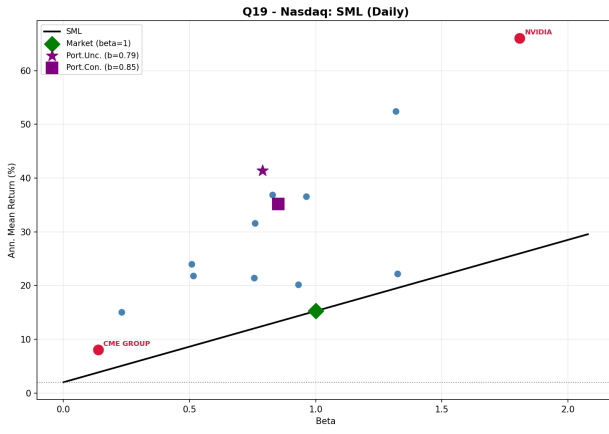
Nasdaq

Nasdaq also shows positive alphas for all securities and portfolios. Cme Group has a small but positive alpha despite its very low beta. Nvidia has the largest alpha in the sample: its high beta implies a high SML return, but its realised return is still much higher. This confirms that market beta alone does not explain the observed performance.

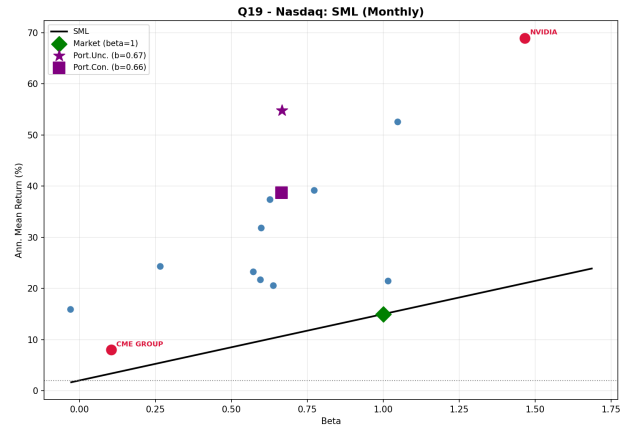
Table 22: Nasdaq-SML

Security / Portfolio	β	Actual $E[R]$ (%)	SML $E[R]$ (%)	α (%)
<i>Daily</i> $r_f = 2.0\%$, $MRP = 13.25\%$				
Cme Group (min β)	0.1381	8.0890	3.8300	4.2590
Nvidia (max β)	1.8077	66.0760	25.9590	40.1170
Portfolio (unconstrained)	0.7894	41.3790	12.4620	28.9170
Portfolio (constrained)	0.8516	35.1590	13.2860	21.8730
<i>Monthly</i> $r_f = 2.0\%$, $MRP = 12.98\%$				
Cme Group (min β)	0.1040	8.0130	3.3500	4.6630
Nvidia (max β)	1.4660	68.9380	21.0250	47.9130
Portfolio (unconstrained)	0.6662	54.7510	10.6450	44.1060
Portfolio (constrained)	0.6643	38.6800	10.6200	28.0600

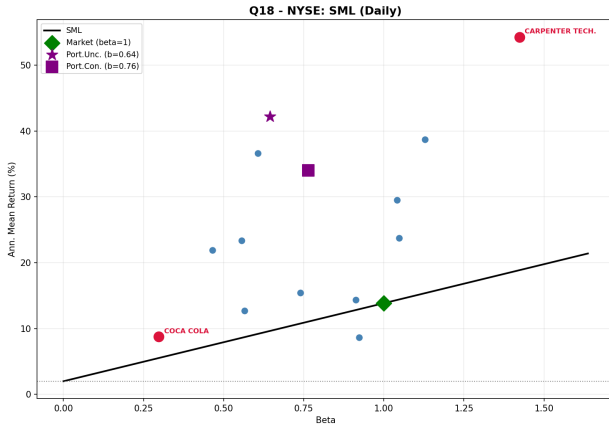
In conclusion, both markets are positioned systematically above the Security Market Line. This suggests that the single-factor CAPM underestimates the realised returns observed in the sample, particularly for high-beta securities and for the unconstrained portfolio allocations. The graphical evidence confirms the same result: securities and portfolios lie above the SML in both markets and at both frequencies, with the largest gaps concentrated in Nasdaq.



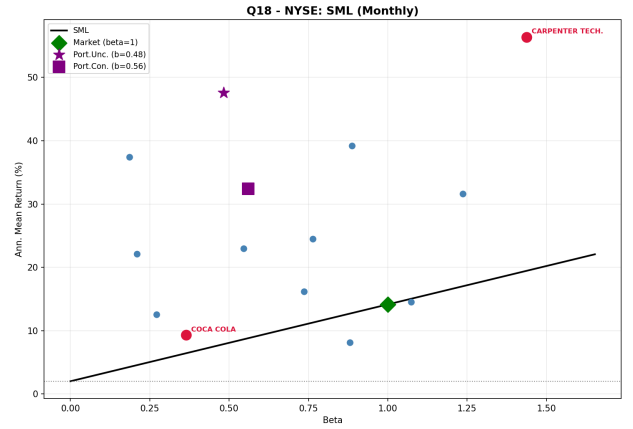
(a) Nasdaq SML daily



(b) Nasdaq SML monthly



(c) NYSE SML daily



(d) NYSE SML monthly

Figure 10: Security Market Line comparison

20-21. Black-Litterman

The Black–Litterman model combines a CAPM-based equilibrium prior with investor views. In this way, it mitigates the instability of sample mean returns by anchoring expected returns to market-implied values and subsequently updating them according to the specified views:

$$\mu_{BL} = [(\tau\Sigma)^{-1} + P^\top\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^\top\Omega^{-1}q],$$

where $\pi = \delta\Sigma w_{eq}$, $\delta = 3.0$, $\tau = 0.05$, and $\Omega = \text{diag}(\tau P\Sigma P^\top)$. The prior component represents equilibrium expected returns, whereas the views incorporate forward-looking information into the return estimates.

Table 23: Black-Litterman Views – NYSE and Nasdaq

Market	Type	View	Target (% p.a.)
NYSE	Absolute	Eli Lilly	35.0000
NYSE	Absolute	Coca Cola	6.0000
NYSE	Relative	Carpenter Tech. – Murphy Oil	15.0000
NYSE	Relative	General Dynamics – Mastercard	5.0000
Nasdaq	Absolute	Nvidia	80.0000
Nasdaq	Absolute	Gilead Sciences	10.0000
Nasdaq	Relative	Broadcom – Cme Group	20.0000
Nasdaq	Relative	Palo Alto Networks – Baker Hughes A	10.0000

NYSE

Eli Lilly receives a 35% absolute view, raising its expected return relative to its low-beta equilibrium prior. Coca Cola receives a moderate 6% view, consistent with its defensive profile. The relative view Carpenter Tech. over Murphy Oil reflects a preference for metals over oil exposure, while General Dynamics over Mastercard reflects stronger expected defence demand relative to digital payments.

Table 24: NYSE – Equilibrium Prior vs BL Posterior

Stock	π_{eq} Daily	μ_{BL} Daily	μ_{BL} Monthly
Carpenter Tech.	16.2290	22.2160	18.2070
Eli Lilly	5.9170	20.6550	18.7570
Caterpillar	10.8120	13.3290	12.3360
Jp Morgan Chase & Co.	8.3870	10.2570	4.7420
Exxon Mobil	8.1740	7.7160	6.4370
Coca Cola	3.0770	5.1380	5.7540
Welltower	5.8550	8.1030	9.2730
Mastercard	7.1760	7.6550	3.6680
Motorola Solutions	6.2150	8.6870	3.0380
Dycom Inds.	13.0670	15.7040	6.7770
General Dynamics	6.2960	9.4310	4.9220
Murphy Oil	15.6820	14.1040	10.4600

Nasdaq

Nvidia receives the strongest view, equal to 80%, reflecting its exceptional growth profile. Gilead Sciences receives a defensive 10% view. The relative views favour Broadcom over Cme Group and Palo Alto Networks over Baker Hughes, consistent with stronger expected growth in semiconductors and cybersecurity.

Table 25: Nasdaq – Equilibrium Prior vs BL Posterior

Stock	π_{eq} Daily	μ_{BL} Daily	μ_{BL} Monthly
Nvidia	22.9330	50.1940	51.8060
Apple	11.2970	19.0740	17.6390
Steel Dynamics	13.2000	19.5260	19.7300
Meta Platforms A	17.0300	28.7250	24.9960
Broadcom	17.6460	29.8880	26.5490
Costco Wholesale	7.2710	11.8830	16.1400
Cme Group	3.0270	5.2210	5.7560
Gilead Sciences	3.8320	7.0160	6.9850
Palo Alto Networks	14.2270	23.3020	21.3030
Baker Hughes A	9.9790	13.8230	12.5810
Interactive Brokers A	13.0900	20.8600	21.6530
Marriott Intl. 'A'	11.1260	17.2460	15.5070

Prior and posterior

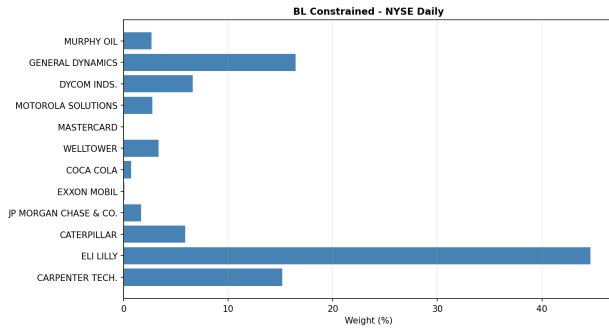
The posterior expected returns adjust in the direction implied by the investor views, while the covariance structure of the assets is preserved. For the NYSE sample, Eli Lilly records a substantial increase as a result of the 35% absolute view, whereas Murphy Oil is revised downward due to the relative view defined against Carpenter Tech. For the Nasdaq sample, Nvidia exhibits the largest revision, while correlated securities such as Apple, Steel Dynamics and Costco also experience upward adjustments through covariance transmission.

Black-Litterman Statistics

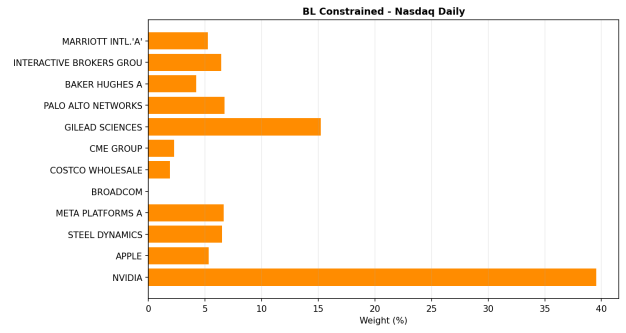
Table 26: Black-Litterman Portfolio Statistics NYSE and Nasdaq

Statistic	NYSE Daily	NYSE Monthly	Nasdaq Daily	Nasdaq Monthly
<i>Unconstrained</i>				
Mean ann. (%)	28.7000	31.5500	34.9700	37.3800
Std ann. (%)	17.5500	16.2800	23.1600	24.0600
Sharpe (ann.)	1.5210	1.8150	1.4230	1.4700
<i>Constrained ($w \geq 0$)</i>				
Mean ann. (%)	33.1500	37.3900	40.5900	43.5800
Std ann. (%)	20.7600	20.2800	27.6100	28.5100
Sharpe (ann.)	1.5000	1.7460	1.3980	1.4580

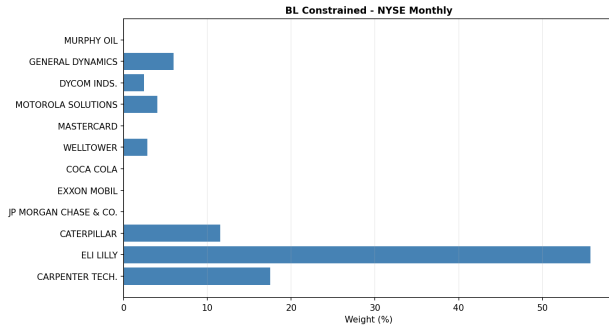
The BL portfolios are consistent across markets and frequencies. In NYSE, the constrained portfolio has higher expected return, while the unconstrained version has slightly higher Sharpe ratios. The same pattern holds for Nasdaq. Nasdaq portfolios are more volatile than NYSE portfolios because of stronger concentration in high-growth, correlated stocks. Monthly frequency generally produces higher Sharpe ratios, indicating that the BL views work better over longer horizons.



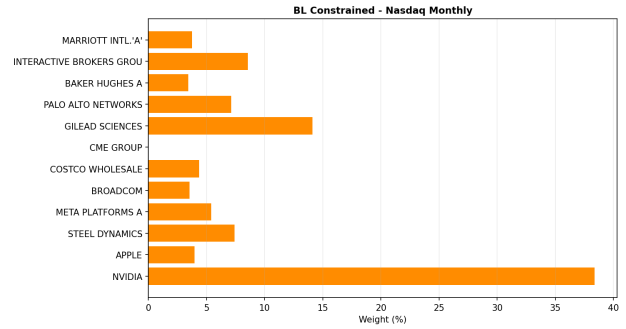
(a) NYSE – Daily



(b) Nasdaq – Daily

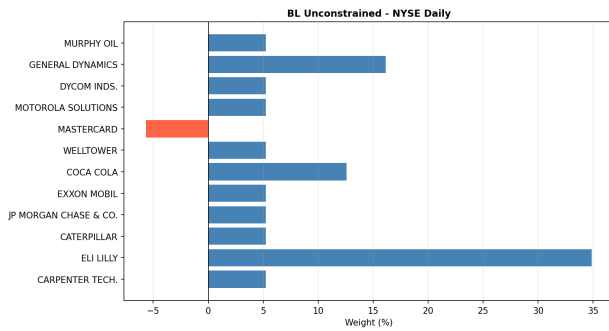


(c) NYSE – Monthly

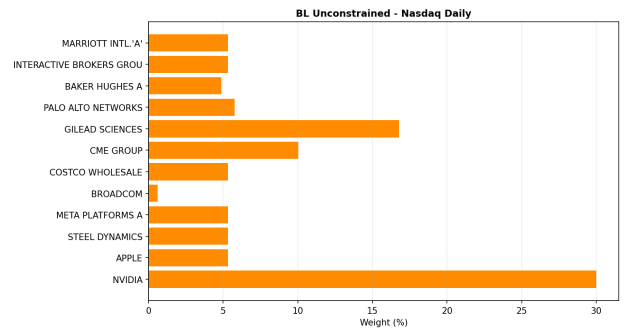


(d) Nasdaq – Monthly

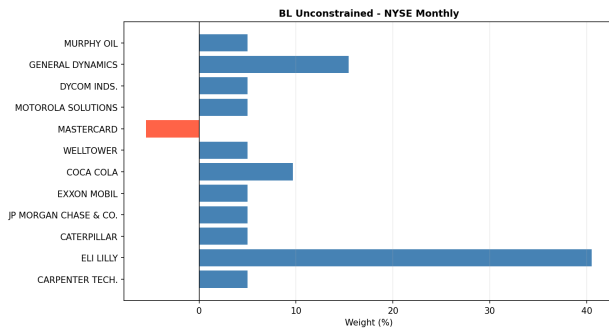
Figure 11: Black-Litterman constrained portfolio weights



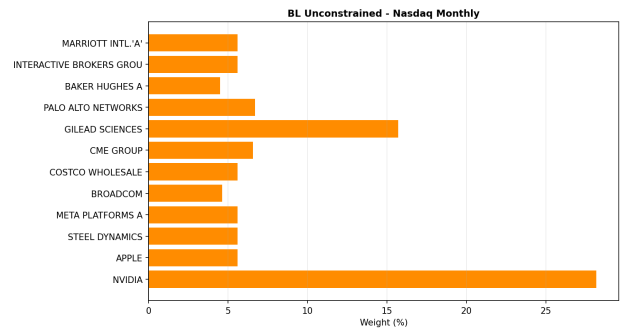
(a) NYSE – Daily



(b) Nasdaq – Daily



(c) NYSE – Monthly



(d) Nasdaq – Monthly

Figure 12: Black-Litterman unconstrained portfolio weights

Pure Bayesian Asset Allocation

The pure Bayesian allocation treats the true mean vector as unknown and estimates it through a prior-posterior update. Unlike Black-Litterman, the prior is entirely data-based. The prior mean is set as

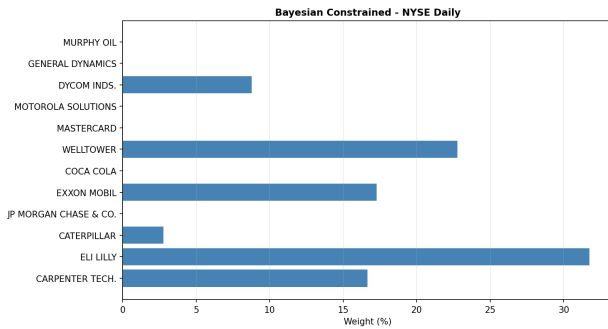
$$\mu_0 = \hat{\mu} + \hat{\sigma},$$

while the prior covariance is $\Sigma_0 = 2\Sigma$. The posterior mean is a precision-weighted average of the prior and sample mean:

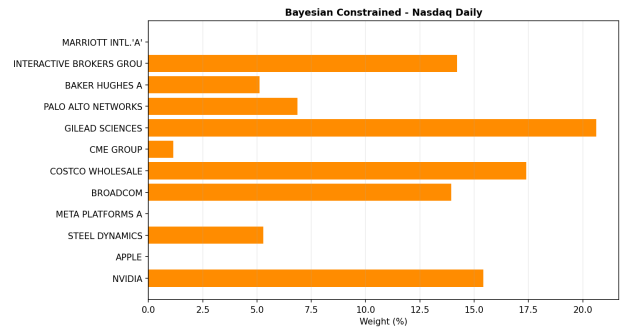
$$\mu_{post} = \Sigma_{post} [\Sigma_0^{-1} \mu_0 + T \Sigma^{-1} \hat{\mu}].$$

Weights are obtained by replacing the sample mean with μ_{post} in the tangency portfolio problem.

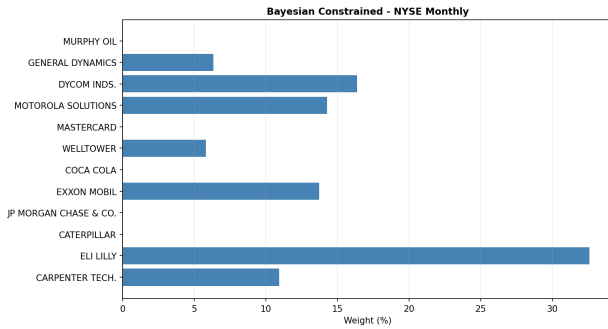
The Bayesian portfolios are more diversified than the Black-Litterman portfolios because the prior does not concentrate views on specific stocks. The main advantage is therefore greater stability and lower sensitivity to extreme sample means.



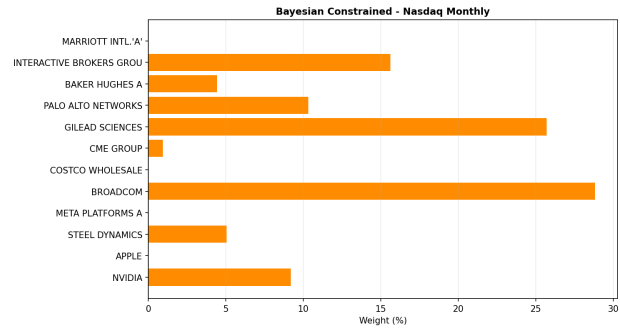
(a) NYSE – Daily



(b) Nasdaq – Daily

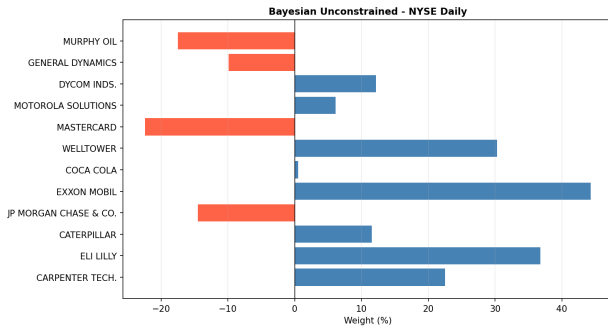


(c) NYSE – Monthly

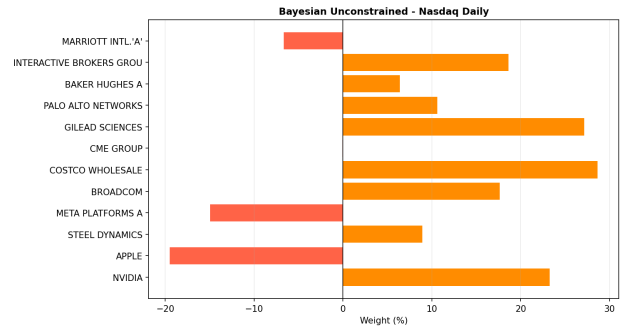


(d) Nasdaq – Monthly

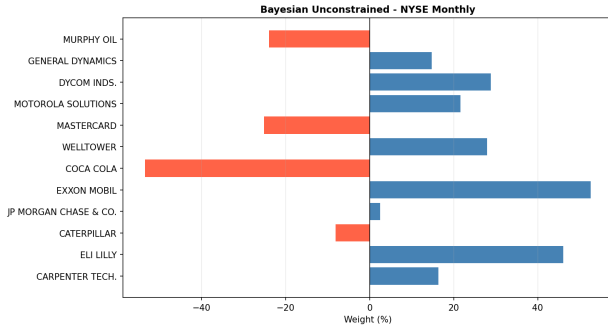
Figure 13: Bayesian constrained portfolio weights



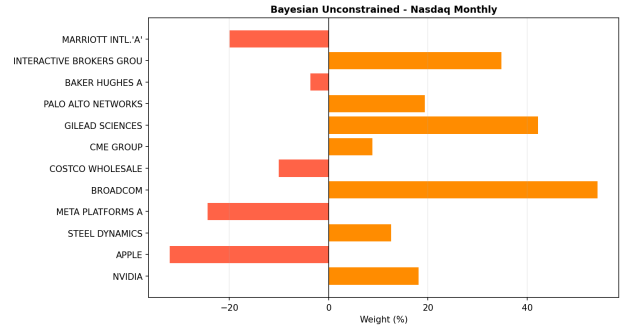
(a) NYSE – Daily



(b) Nasdaq – Daily



(c) NYSE – Monthly



(d) Nasdaq – Monthly

Figure 14: Bayesian unconstrained portfolio weights

22-23. GMVP Model

The Global Minimum Variance Portfolio minimises portfolio variance without relying on expected returns. In the unconstrained case, the optimal allocation is given by:

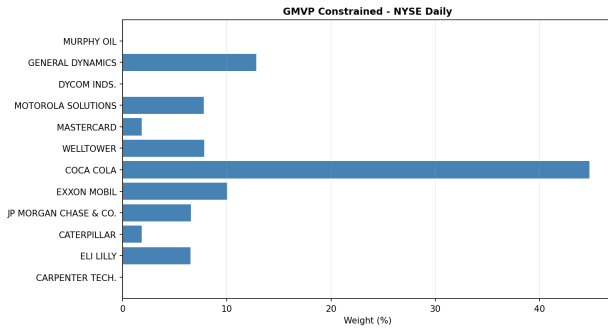
$$w^* = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}, \quad \sigma_{mv}^2 = \frac{1}{c}, \quad c = \mathbf{1}^\top \Sigma^{-1}\mathbf{1}, \quad \mu_{mv} = \frac{b}{c}, \quad b = \mathbf{1}^\top \Sigma^{-1}\mu.$$

Since this portfolio depends only on the covariance matrix, the GMVP is relatively robust to estimation errors in expected returns. The constrained version, where $w \geq 0$, is instead obtained numerically through the SLSQP algorithm.

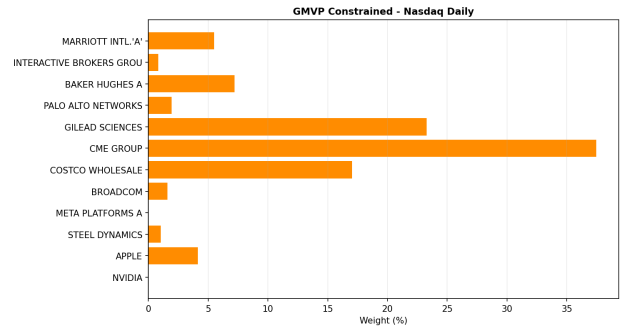
The NYSE GMVP exhibits lower volatility than the Nasdaq GMVP at both daily and monthly frequencies, with the monthly NYSE allocation achieving the lowest volatility in the overall analysis. This outcome is mainly driven by large weights assigned to defensive and low-volatility securities, such as Coca Cola, General Dynamics and Welltower. By contrast, the Nasdaq universe is more difficult to diversify, leading the optimiser to concentrate primarily on Cme Group and Costco Wholesale, while assigning only limited weights to the most volatile technology stocks. In some cases, the constrained GMVP also delivers a higher Sharpe ratio, as the exclusion of short positions prevents the expected return from being reduced too strongly.

Table 27: GMVP – Analytical Scalars and Portfolio Statistics

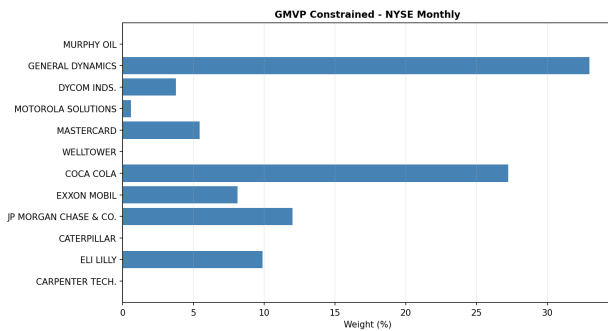
Metric	NYSE Daily	NYSE Monthly	Nasdaq Daily	Nasdaq Monthly
$c = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$	16 103.3800	1357.2600	13 279.5800	686.6700
$b = \mathbf{1}^\top \Sigma^{-1} \mu$	9.2500	12.7700	8.0700	7.5000
σ_{mv} ann. (%)	12.5100	9.4000	13.7800	13.2200
μ_{mv} ann. (%)	14.4800	11.2900	15.3100	13.1000
<i>Unconstrained</i>				
Mean ann. (%)	14.4800	11.2900	15.3100	13.1000
Std ann. (%)	12.5100	9.4000	13.7800	13.2200
Sharpe (ann.)	0.9980	0.9880	0.9660	0.8400
<i>Constrained ($w \geq 0$)</i>				
Mean ann. (%)	14.8100	15.9100	16.1700	16.8900
Std ann. (%)	12.8000	10.7100	13.8400	13.7600
Sharpe (ann.)	1.0010	1.2990	1.0240	1.0820



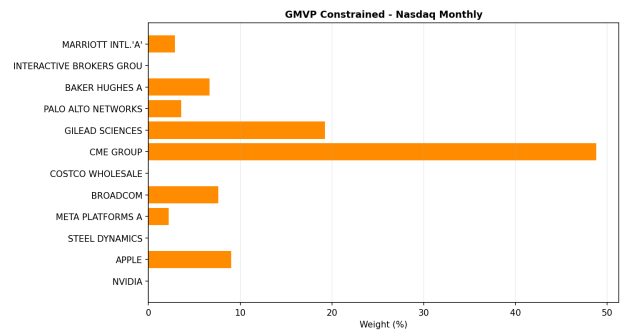
(a) NYSE – Daily



(b) Nasdaq – Daily

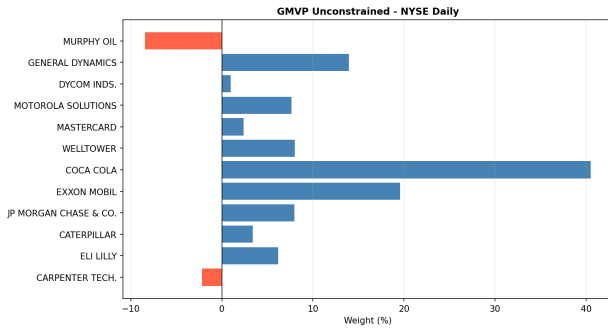


(c) NYSE – Monthly

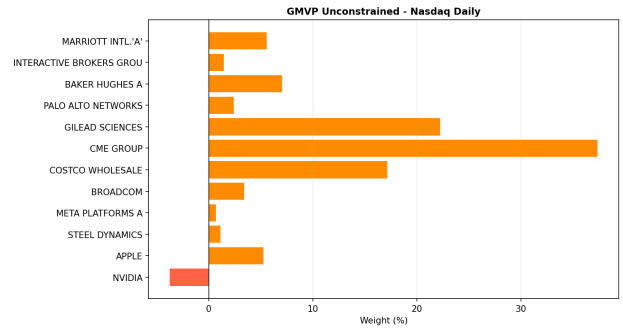


(d) Nasdaq – Monthly

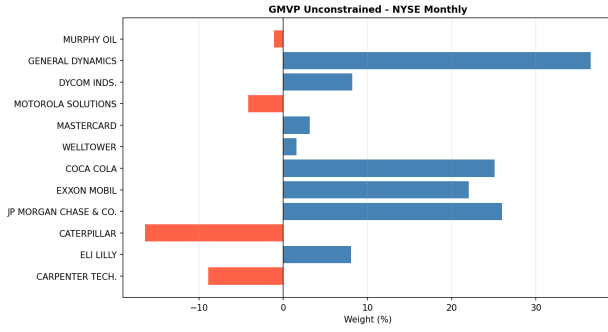
Figure 15: GMVP constrained portfolio weights



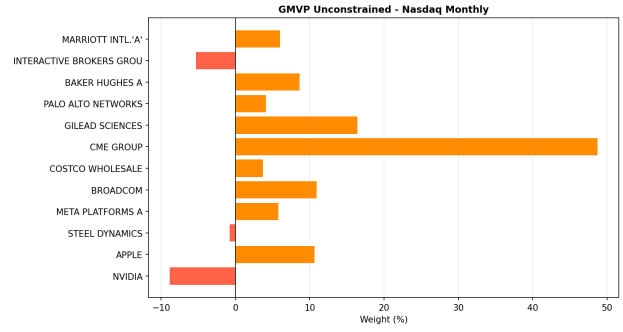
(a) NYSE – Daily



(b) Nasdaq – Daily



(c) NYSE – Monthly



(d) Nasdaq – Monthly

Figure 16: GMVP unconstrained portfolio weights

24. Linear Combinations

The four portfolio models rely on different assumptions. MV trusts the sample mean, BL combines equilibrium returns with views, Bayesian allocation shrinks sample estimates toward a data-based prior and GMVP ignores expected returns completely. Combining them reduces model risk in the same way asset diversification reduces idiosyncratic risk. The combined portfolio is the following:

$$w_c = \lambda_{MV} w_{MV} + \lambda_{BL} w_{BL} + \lambda_{Bayes} w_{Bayes} + \lambda_{GMVP} w_{GMVP}$$

with four weighting schemes.

Table 28: Combination Schemes

Strategy	λ_{MV}	λ_{BL}	λ_{Bayes}	λ_{GMVP}
Equal (25/25/25/25)	0.25	0.25	0.25	0.25
Risk-focused (10/20/30/40)	0.10	0.20	0.30	0.40
Return-focused (40/40/10/10)	0.40	0.40	0.10	0.10
Bayes-dom. (15/35/35/15)	0.15	0.35	0.35	0.15

The schemes range from conservative to aggressive, where equal weighting is the most neutral, the risk-focused scheme gives more weight to Bayesian and GMVP portfolios, the return-focused scheme emphasises MV and BL, while the Bayes-dominant scheme combines Black-Litterman views with Bayesian shrinkage.

Pure Portfolio Statistics

Table 29: Pure Portfolio Statistics – Nasdaq

Method	Nasdaq Daily					Nasdaq Monthly				
	$E[R]\%$	$\sigma\%$	Skew	Kurt	SR	$E[R]\%$	$\sigma\%$	Skew	Kurt	SR
<i>Constrained ($w \geq 0$)</i>										
MV	35.16	21.12	0.206	5.440	1.570	38.68	21.85	0.564	0.890	1.679
BL	40.59	27.61	0.349	4.362	1.398	43.58	28.51	0.558	0.952	1.458
Bayes	35.15	21.11	0.206	5.440	1.570	38.31	21.63	0.560	0.882	1.679
GMVP	16.17	13.84	-0.272	3.802	1.024	16.89	13.76	-0.154	0.717	1.082
<i>Unconstrained</i>										
MV	41.38	23.68	0.051	3.509	1.663	54.75	28.80	1.151	2.369	1.832
BL	34.97	23.16	0.266	4.411	1.423	37.38	24.06	0.422	0.991	1.470
Bayes	41.36	23.67	0.051	3.509	1.663	53.88	28.32	1.143	2.353	1.832
GMVP	15.31	13.78	-0.278	3.646	0.966	13.10	13.22	-0.019	0.813	0.840

Table 30: Pure Portfolio Statistics – NYSE

Method	NYSE Daily					NYSE Monthly				
	$E[R]\%$	$\sigma\%$	Skew	Kurt	SR	$E[R]\%$	$\sigma\%$	Skew	Kurt	SR
<i>Constrained ($w \geq 0$)</i>										
MV	34.00	19.32	0.109	2.632	1.657	32.44	15.41	-0.130	-0.608	1.976
BL	33.15	20.30	0.241	4.339	1.500	37.39	19.88	0.064	-0.159	1.746
Bayes	34.00	19.31	0.109	2.632	1.657	32.31	15.34	-0.130	-0.609	1.976
GMVP	14.81	12.80	-0.362	4.355	1.001	15.91	10.71	0.340	1.275	1.299
<i>Unconstrained</i>										
MV	42.18	22.44	0.266	1.680	1.791	47.61	20.81	-0.013	-0.830	2.191
BL	27.08	17.26	0.164	3.966	1.453	28.87	15.60	0.068	-0.240	1.722
Bayes	42.16	22.42	0.266	1.680	1.791	46.89	20.49	-0.015	-0.834	2.191
GMVP	14.48	12.51	-0.373	3.617	0.998	11.29	9.40	0.167	0.106	0.989

The pure portfolio statistics show the role of each model. MV and Bayesian are very close, especially in constrained Nasdaq, confirming that Bayesian shrinkage improves stability without materially reducing performance. BL generally earns higher returns but also carries higher volatility. GMVP has the lowest risk and return, making it mainly useful as a stabilising component. Removing constraints often improves Sharpe ratios, especially for NYSE monthly, but the effect depends on how short positions interact with expected returns.

Combination Results

Equal weighting (25/25/25/25).

Equal weighting is the most agnostic scheme and performs very well across markets. It achieves strong Sharpe ratios without relying on any single model, confirming the benefit of model diversification.

Risk-focused (10/20/30/40).

This scheme gives 70% weight to Bayesian and GMVP portfolios. It delivers the lowest volatility and expected return, and is therefore appropriate mainly for investors with strict risk or drawdown constraints.

Return-focused (40/40/10/10).

This is the most aggressive scheme because it concentrates on MV and BL. It usually produces the highest expected returns and strong positive skewness, but also accepts higher volatility and greater exposure to high-growth names.

Bayes-dominant (15/35/35/15).

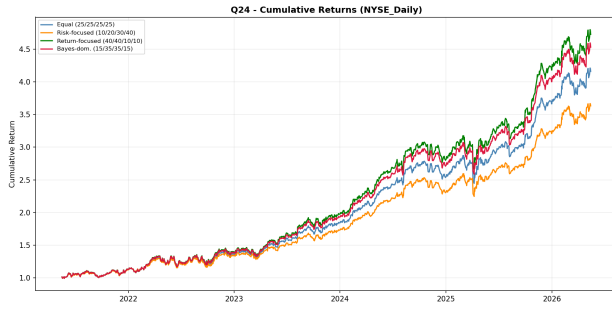
This scheme combines BL and Bayesian allocation. Its results are close to the return-focused scheme, showing that forward-looking views and statistical shrinkage provide complementary information.

Table 31: Combination Results – Nasdaq

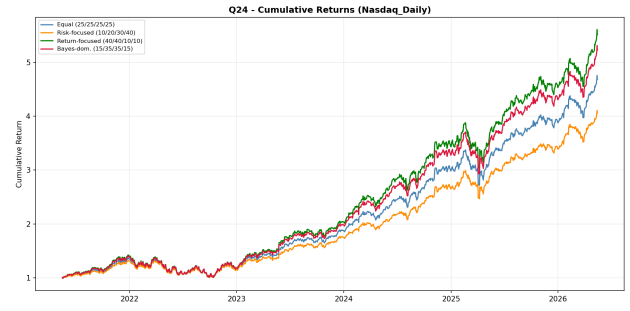
Strategy	Nasdaq Daily					Nasdaq Monthly				
	$E[R]\%$	$\sigma\%$	Skew	Kurt	SR	$E[R]\%$	$\sigma\%$	Skew	Kurt	SR
<i>Constrained ($w \geq 0$)</i>										
Equal (25/25/25/25)	31.77	19.55	0.157	4.996	1.522	34.37	20.10	0.416	0.934	1.610
Risk-focused (10/20/30/40)	28.64	17.78	0.077	4.930	1.498	30.83	18.28	0.295	0.844	1.577
Return-focused (40/40/10/10)	35.43	22.11	0.235	4.923	1.512	38.43	22.71	0.508	1.025	1.604
Bayes-dom. (15/35/35/15)	34.21	21.23	0.212	4.947	1.517	37.00	21.77	0.479	1.004	1.607
<i>Unconstrained</i>										
Equal (25/25/25/25)	33.26	19.34	0.056	3.902	1.616	39.78	21.32	0.866	1.850	1.772
Risk-focused (10/20/30/40)	29.67	17.54	-0.007	4.032	1.577	34.36	18.78	0.665	1.452	1.723
Return-focused (40/40/10/10)	36.21	21.21	0.128	4.010	1.613	43.55	23.53	0.933	1.984	1.766
Bayes-dom. (15/35/35/15)	35.22	20.56	0.105	3.974	1.616	42.12	22.69	0.908	1.944	1.768

Table 32: Combination Results – NYSE

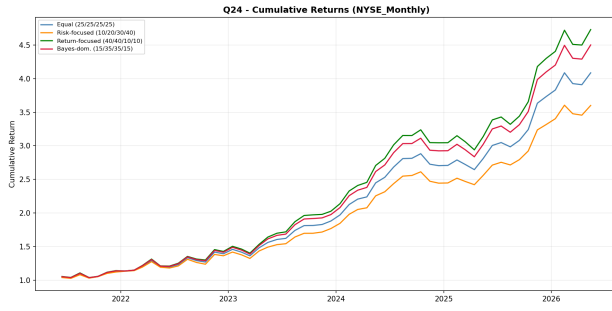
Strategy	NYSE Daily					NYSE Monthly				
	$E[R]\%$	$\sigma\%$	Skew	Kurt	SR	$E[R]\%$	$\sigma\%$	Skew	Kurt	SR
<i>Constrained ($w \geq 0$)</i>										
Equal (25/25/25/25)	28.44	16.87	0.060	3.340	1.568	28.96	14.45	-0.053	-0.404	1.866
Risk-focused (10/20/30/40)	25.72	15.59	-0.025	3.493	1.521	26.34	13.40	-0.001	-0.239	1.817
Return-focused (40/40/10/10)	30.87	18.36	0.143	3.435	1.573	31.86	15.99	-0.073	-0.450	1.867
Bayes-dom. (15/35/35/15)	30.06	17.84	0.118	3.396	1.573	30.87	15.45	-0.068	-0.443	1.869
<i>Unconstrained</i>										
Equal (25/25/25/25)	31.48	17.07	0.145	2.226	1.727	33.66	14.96	-0.030	-0.879	2.116
Risk-focused (10/20/30/40)	28.08	15.58	0.039	2.344	1.673	29.12	13.21	-0.040	-0.835	2.053
Return-focused (40/40/10/10)	33.37	18.14	0.217	2.453	1.730	36.41	16.29	-0.006	-0.842	2.113
Bayes-dom. (15/35/35/15)	32.73	17.77	0.194	2.375	1.730	35.35	15.77	-0.014	-0.858	2.114



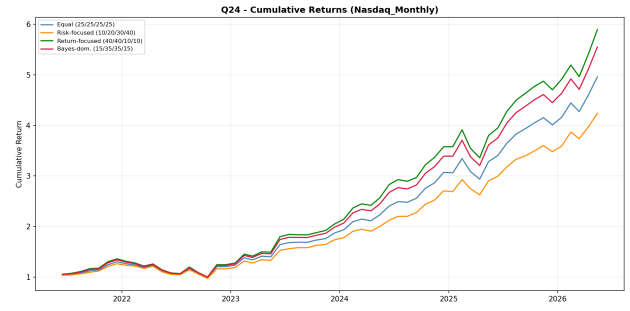
(a) NYSE – Daily (constrained)



(b) Nasdaq – Daily (constrained)

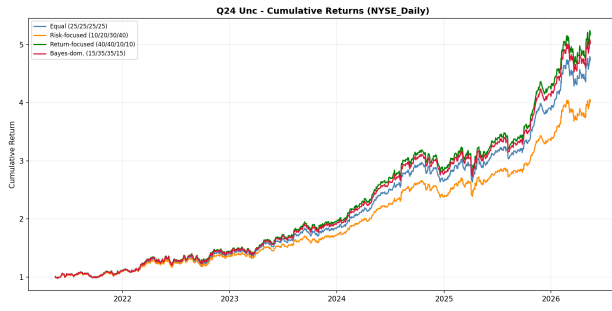


(c) NYSE – Monthly (constrained)

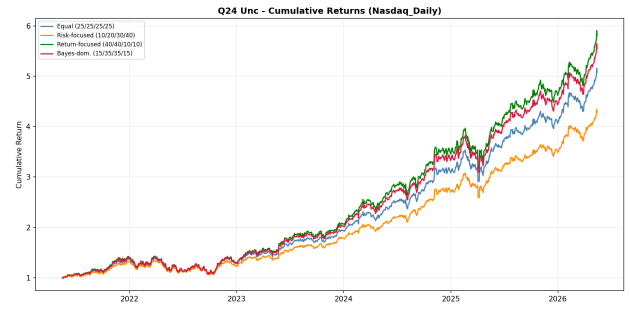


(d) Nasdaq – Monthly (constrained)

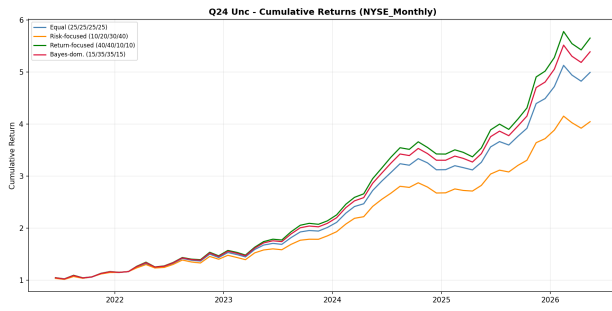
Figure 17: Cumulative returns – Linear combinations (constrained)



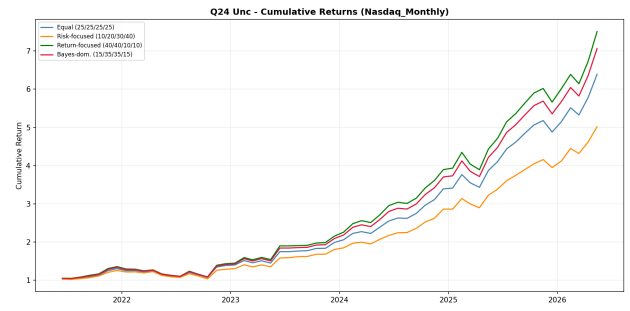
(a) NYSE – Daily



(b) Nasdaq – Daily



(c) NYSE – Monthly



(d) Nasdaq – Monthly

Figure 18: Cumulative returns – Linear combinations, unconstrained (Daily and Monthly)

Cross-cutting Observations

Kurtosis

Nasdaq daily combinations continue to display positive excess kurtosis, indicating that they remain exposed to the fat-tailed behavior typically associated with technology stocks. In the unconstrained case, kurtosis is reduced because the possibility of short selling helps lower portfolio concentration. By contrast, NYSE monthly combinations exhibit negative excess kurtosis, which is consistent with broader sector diversification and a distribution characterized by thinner tails.

Constrained vs Unconstrained

Removing the no-shorting constraint generally leads to higher Sharpe ratios and helps reduce portfolio concentration, as the allocation is no longer restricted to long-only positions. This effect is particularly pronounced for the NYSE monthly case, where the improvement in risk-adjusted performance is strongest. However, allowing unconstrained allocations may also increase exposure to securities with more extreme return behavior, especially in the Nasdaq portfolios.

Daily vs Monthly

Monthly portfolios generally exhibit higher Sharpe ratios than daily portfolios, suggesting stronger risk-adjusted performance at the monthly frequency. This result is consistent with the fact that monthly returns are less influenced by short-run noise, while covariance estimates tend to be more stable over longer intervals. In addition, the Black–Litterman views are naturally better aligned with a medium-term investment horizon. Overall, these considerations support the use of monthly rebalancing rather than daily rebalancing.